
Low-rank Covariance Matrix Estimation from Quadratic Measurements

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Abstract

Covariance matrices, as fundamental second-order statistics, play a crucial role in guiding learning and inference processes in various artificial intelligence applications, including collaborative filtering and principal component analysis. In practical scenarios, where data is often dynamic and resource-constrained, it presents a significant challenge for accurately estimating the covariance matrix from a single pass over the data with compressive measurements. In this paper, we consider the problem of recovering low-rank covariance matrices from quadratic measurements combined with nonconvex regularization. We introduce a proximal gradient homotopy method to solve the proposed estimator. Theoretical analysis shows that our method achieves superior statistical rates compared to estimators based on the convex nuclear norm, and under mild assumptions, it attains the oracle rate. These theoretical improvements are supported by extensive numerical experiments on both synthetic data and real-world datasets.

1 INTRODUCTION

The covariance matrix, a fundamental tool for quantifying linear correlations among multiple random variables, is central to a wide range of statistical analyses and data-driven applications Loh [1988], Pourahmadi [2013], Bose and Bhattacharjee [2018], Zagidullina [2021]. Its significance spans across numerous domains Szatrowski [1980], Hung and Kaveh [1987], Yao et al. [2015], Sihag et al. [2022], including principal component analysis Jolliffe [2002], linear and quadratic discriminant analysis Hastie et al. [2009], portfolio optimization Markowitz [1952], etc. Consider a zero-mean data stream $\{\mathbf{x}_t \in \mathbb{R}^d\}_{t \in \mathcal{T}}$ with covariance matrix Σ^* . A common approach to estimating the

covariance matrix is the sample covariance matrix (SCM) $\mathbf{S} = \frac{1}{|\mathcal{T}|} \sum_{t=1}^{|\mathcal{T}|} \mathbf{x}_t \mathbf{x}_t^\top$ Wishart [1928], Richard et al. [2012], Shikhaliev et al. [2019], Wei and Zhao [2023]. However, the computation of SCM typically requires sufficient data samples Vershynin [2012], which may be computationally expensive and resource-intensive Johnstone [2007], Karoui [2008]. A promising alternative is to estimate the covariance matrix from compressed measurements, circumventing the need for complete data. A practical approach is to project the data $\mathbf{x}$ onto a lower-dimensional representation $\mathbf{a}^\top \mathbf{x}$, where $\mathbf{a}$ is a sketching vector. The covariance matrix can then be estimated from these compressed measurements Ariananda and Leus [2011, 2012], Ariananda et al. [2013], Romero et al. [2015], Chen et al. [2015], Zhong et al. [2015].

In this paper, we investigate a quadratic measurement model Chen et al. [2015], Zhong et al. [2015], Li et al. [2021] taking the form:

$$y_i = \frac{1}{|\mathcal{T}|} \sum_{t=1}^{|\mathcal{T}|} (\mathbf{a}_i^\top \mathbf{x}_t)^2, \quad i = 1, 2, \dots, m$$

where $\{\mathbf{a}_i\}_{i=1}^m$ are predefined sketching vectors. As the number of data instances increases, the measurement y_i converges to $\mathbb{E}[(\mathbf{a}_i^\top \mathbf{x}_t)^2] = \mathbf{a}_i^\top \Sigma^* \mathbf{a}_i$. This framework offers significant advantages for high-dimensional data, as it substantially reduces both the storage and computational costs Chen et al. [2015, 2017]. However, a key challenge arises from the ill-posed nature of the problem: the number of available measurements is insufficient relative to the degrees of freedom of the covariance matrix. To address this issue, leveraging structural properties of the covariance matrix—such as sparsity Tibshirani [1996] and low-rankness Rohde and Tsybakov [2011], Lounici [2014]—has been proven highly effective.

We consider the problem of recovering a low-rank covariance matrix from compressed measurements, which are prevalent in various applications, including phase retrieval Jaganathan et al. [2012], Candes et al. [2013], Netrapalli et al. [2013], Beck and Eldar [2013], Candès and Li [2014],

Waldspurger et al. [2015], PhaseLift Candes et al. [2013], mixed linear regression Zhong et al. [2016], quantum state tomography Cramer et al. [2010], among others. Two primary methodological paradigms have emerged for low-rank matrix recovery: penalty-based methods and factorization-based methods. Factorization-based techniques decompose the matrix into two low-rank factors (e.g., $\Sigma^* = \mathbf{U}\mathbf{U}^\top$ with $\mathbf{U} \in \mathbb{R}^{d \times r}$ ($r \ll d$)). While these methods are often computationally efficient, they are prone to overfitting, particularly when data are non-sparse Koren et al. [2009]. Additionally, they require prior knowledge of the rank r , which is often unknown in practice. Moreover, the resulting optimization problem is generally nonconvex, complicating the theoretical error analysis. Existing theoretical guarantees predominantly focus on bounding the error of the factorized components but fail to offer direct recovery error relative to the ground truth Li et al. [2021]. Furthermore, these methods demand careful initialization—such as spectral initialization Li et al. [2021]—to ensure convergence to meaningful solutions. In contrast, penalty-based methods offer a solid theoretical foundation for low-rank matrix recovery Rohde and Tsybakov [2011], Negahban and Wainwright [2011], Lounici [2014], Gui et al. [2016]. One direct approach is through rank minimization to achieve accurate recovery of a low-rank matrix. However, this approach is computationally intractable, leading to the development of surrogate loss functions, such as the nuclear norm Candès and Tao [2010], Schatten- p norm Nie et al. [2012], and weighted nuclear norm Chen et al. [2013], among others. The nuclear norm has been praised for its close convex approximation to the matrix rank, facilitating global optimality Negahban and Wainwright [2011], Koltchinskii et al. [2011]. A more advanced variant, the weighted nuclear norm, assigns different weights to singular values, enhancing stability and accuracy over the conventional nuclear norm penalty Huang et al. [2020]. Despite their effectiveness, these ℓ_1 -homotopy penalties lead to over-penalization and suboptimal feature selection Zhang [2010b], Candes et al. [2008]. To overcome this problem, several nonconvex penalties, such as the smoothly clipped absolute deviation (SCAD) penalty Fan and Li [2001] and the minimax concave penalty (MCP) Zhang [2010a], have been proposed and demonstrated promising empirical results Liu et al. [2013], Wang et al. [2014], Gui et al. [2016].

Contributions. We introduce a novel approach for estimating low-rank covariance matrices from the quadratic measurement model, leveraging a non-convex penalty to produce unbiased estimates. To efficiently solve the resulting estimator, we propose a proximal gradient homotopy method. Our theoretical analysis demonstrates that the proposed estimator achieves superior statistical convergence rates compared to traditional nuclear norm-penalized estimator. Under mild assumptions, we show that our method possesses the oracle property, enabling accurate recovery of the true rank of the covariance matrix. The efficacy of

the proposed method is further validated through extensive simulation studies, which highlight its superior performance relative to existing methods.

2 LOW-RANK COVARIANCE ESTIMATION FROM QUADRATIC MEASUREMENT MODEL

In this section, we present a unified framework for low-rank covariance matrix estimation within the quadratic measurement model. We then introduce our estimator, incorporating a nonconvex penalty, and demonstrate its applicability through several practical applications.

2.1 THE QUADRATIC MEASUREMENT MODEL

Consider a set of independent observation streams $\{\mathbf{x}_t \in \mathbb{R}^d\}_{t \in \mathcal{T}}$, where each $\mathbf{x}_t$ is drawn from a zero-mean vector $\mathbf{x}$ with covariance matrix Σ^* . The measurements are modeled defined as follows¹:

$$y_i = \mathbf{a}_i^\top \mathbf{S} \mathbf{a}_i + \eta_i, \quad i = 1, \dots, m, \quad (1)$$

where $\{y_i\}_{i=1}^m$ represent the observed measurements, $\mathbf{S} = \frac{1}{|\mathcal{T}|} \sum_{t=1}^{|\mathcal{T}|} \mathbf{x}_t \mathbf{x}_t^\top$ is the sample covariance matrix, $\{\mathbf{a}_i\}_{i=1}^m$ are the sensing vectors, and $\{\eta_i\}_{i=1}^m$ are the noise terms. We assume that the noise term η_i 's ($1 \leq i \leq m$) are independent and follow a sub-Gaussian distribution with zero mean and variance proxy δ^2 . For convenience, let $\mathbf{y} := [y_1, \dots, y_m]^\top$ be the measurement vector, and $\boldsymbol{\eta} := [\eta_1, \dots, \eta_m]^\top$ the noise vector. We introduce the linear operator $\mathcal{A}(\cdot) : \mathbb{R}^{d \times d} \mapsto \mathbb{R}^m$, defined by $\mathcal{A}(\mathbf{S}) = [\langle \mathbf{a}_1 \mathbf{a}_1^\top, \mathbf{S} \rangle, \dots, \langle \mathbf{a}_m \mathbf{a}_m^\top, \mathbf{S} \rangle]$ for any $\mathbf{S} \in \mathbb{R}_+^{d \times d}$. With these definitions, the measurement model can be compactly expressed as

$$\mathbf{y} = \mathcal{A}(\mathbf{S}) + \boldsymbol{\eta}.$$

This quadratic measurement model, previously explored by Candes et al. [2013], Chen et al. [2015], offers a simpler implementation and improved computational efficiency compared to models that rely on full-rank measurement matrices with independently and identically distributed entries.

2.2 PROPOSED NONCONVEX ESTIMATOR

Given a collection of m samples $\{y_i, \mathbf{a}_i\}_{i=1}^m$, we aim to estimate the covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$ and propose the following regularized least squares optimization estimator:

$$\min_{\Sigma \succeq 0} \left\{ \frac{1}{2m} \|\mathbf{y} - \mathcal{A}(\Sigma)\|_2^2 + \sum_{i=1}^d p_\lambda(\sigma_i(\Sigma)) \right\}, \quad (2)$$

¹Given finite samples, the sample covariance matrix can be expressed as $\mathbf{S} = \Sigma^* + \mathbf{E}$, where $\mathbf{E}$ denotes the bias term.

where $\sigma_i(\Sigma)$ denotes the i -th largest eigenvalue of Σ , and $p_\lambda(\cdot)$ is a non-convex penalty function governed by a regularization parameter $\lambda > 0$. The function $p_\lambda(\cdot)$ satisfies specific conditions, as detailed in Assumption 1.

Assumption 1. *The function $p_\lambda(t)$ is symmetric around zero, and can be decomposed as $p_\lambda(t) = \lambda|t| + q_\lambda(t)$, where the functions $p_\lambda(t)$ and $q_\lambda(t)$ satisfy the following conditions:*

- (a) For all $t_1 \geq t_2 \geq 0$, the derivative satisfies $0 \leq p'_\lambda(t_1) \leq p'_\lambda(t_2) \leq \lambda$ and $\lim_{t \rightarrow 0} p'_\lambda(t) = \lambda$;
- (b) $q'_\lambda(t)$ is monotonic and Lipschitz continuous. Specifically, for $t_1 \geq t$, there exists a constant $\zeta^- \geq 0$ such that $q'_\lambda(t_1) - q'_\lambda(t) \geq -\zeta^-(t_1 - t)$.
- (c) $p_\lambda(t)$ is non-decreasing with $p_\lambda(0) = 0$ and is differentiable almost everywhere on $(0, +\infty)$;
- (d) There exists an $\alpha > 0$ such that $p'_\lambda(t) = 0$ for $t \geq \alpha$;

These conditions align with the desired properties for an estimator, which include unbiasedness, sparsity, and continuity Fan and Li [2001]. Several penalty functions satisfying Assumption 1 have been proposed. Below, we present a few representative examples:

- **Smooth clipped absolute deviation (SCAD) penalty:** This penalty, due to Fan and Li [2001], is given by

$$p'_\lambda(t) := \begin{cases} \lambda, & \text{for } 0 < t \leq \lambda, \\ \frac{b\lambda - t}{b-1}, & \text{for } \lambda \leq t \leq b\lambda, \\ 0, & \text{for } t \geq b\lambda, \end{cases}$$

where $b > 2$ is an additional tuning parameter. Fan and Li suggested $b = 3.7$ for cases where the variable dimension is less than 100 through a Bayesian rationale.

- **Minimax concave penalty (MCP):** This penalty, due to Zhang [2010a], is defined as follows:

$$p_\lambda(t) := \text{sign}(t) \lambda \cdot \int_0^{|t|} \left(1 - \frac{z}{\lambda b}\right)_+ dz$$

for some constant $b > 0$.

2.3 APPLICATIONS

The quadratic measurements, as described in (1), are widely employed across numerous domains, ranging from traffic data monitoring Shao et al. [2014], array signal processing Ottersten et al. [1998] and collaborative filtering Yu et al. [2009], Zhang et al. [2010] to metric learning Sivalingam et al. [2009], Bellet et al. [2015]. Here, we present a brief review of two representative examples: PhaseLift and spectrum estimation.

- **PhaseLift.** PhaseLift is a crucial technique for addressing the phase retrieval problem, which seeks to estimate an unknown signal $\mathbf{x}$ from intensity measurements. These intensity measurements are often modeled by the quadratic measurements of the form $y_i = (\mathbf{a}_i^\top \mathbf{x})^2$, where $\mathbf{a}_i$ is the measurement vector. This problem has numerous applications in fields such as X-ray crystallography, optical imaging, and astronomy. PhaseLift tackles this challenge by lifting the problem into a high-dimensional space, transitioning the task of recovering the unknown signal $\mathbf{x}$ into the task of recovering the matrix $\mathbf{X} = \mathbf{x}\mathbf{x}^H$, where $\mathbf{x}^H$ denotes the conjugate transpose of $\mathbf{x}$, and $\mathbf{X}$ is a rank-1 complex Hermitian matrix. This lifting technique converts the original non-linear problem into a tractable linear problem through convex optimization. For a comprehensive discussion on the equivalence conditions between the original and lifted formulations, we refer readers to Candes et al. [2013]. Furthermore, this approach is closely related to problems such as mixed linear regression, where the observed data are generated by one of two unknown regressors; see Chen et al. [2014] for a precise formulation.

- **Spectrum estimation.** Spectrum estimation is a fundamental technique for analyzing the frequency characteristics of signals within stochastic processes, with the primary goal of accurately recovering the power spectrum (or power spectral density) of a signal, which is intrinsically linked to its autocorrelation function. Specifically, spectrum sensing involves estimating the power spectrum of a wide-sense stationary analog signal $x(t)$. This process utilizes a sampling architecture consisting of M parallel branches. Each branch modulates the input signal $x(t)$ with a complex-valued periodic waveform $p_i(t)$ that has a period of NT . The modulated signal is then processed through an integrate-and-dump circuit operating with the same period NT , effectively reducing the sampling rate to $\frac{1}{N}$ of the Nyquist rate. The output of the i -th branch at the k -th sampling interval is given by

$$y_i[k] = \frac{1}{NT} \int_{kNT}^{(k+1)NT} p_i(t)^\top x(t) dt.$$

By leveraging this relationship, the goal is to reconstruct the power spectrum of $x(t)$ using the collected samples $\{y_i[k]\}_{i,k}$. Ariananda and Leus [2012]. This approach facilitates efficient spectrum estimation by exploiting the multibranch sampling architecture and the inherent properties of wide-sense stationary signals.

3 OPTIMIZATION ALGORITHM

We propose solving the problem (2) using a proximal gradient algorithm. Under Assumption 1, the penalty term

$\sum_{i=1}^d p_\lambda(\sigma_i(\Sigma))$ can be written as a combination of $\lambda \|\Sigma\|_*$ and $Q_\lambda(\Sigma) = \sum_{i=1}^d q_\lambda(\sigma_i(\Sigma))$, where $Q_\lambda(\Sigma)$ represent the concave component. Define $\tilde{f}(\Sigma) = f(\Sigma) + Q_\lambda(\Sigma)$, where $f(\Sigma) = \frac{1}{2m} \|\mathbf{y} - \mathcal{A}(\Sigma)\|_2^2$. The optimization problem is then reformulated as:

$$\min_{\Sigma \succeq 0} h(\Sigma), \text{ where } h(\Sigma) = \tilde{f}(\Sigma) + \lambda \|\Sigma\|_*. \quad (3)$$

To estimate $\hat{\Sigma}$ from problem (3), we develop a proximal gradient method with backtracking line search Rolfs et al. [2012]. Starting from $\Sigma_0 = \mathbf{0}$ for simplicity, we generate a sequence $\{\Sigma_k\}_{k \geq 0}$ via the proximal gradient update as described below.

At the $(k+1)$ -th iteration, for a given regularization parameter λ_{k+1} , we construct an isotropic quadratic approximation of $\tilde{f}(\Sigma)$ at Σ_k as follows:

$$\begin{aligned} \tilde{g}(\Sigma | \Sigma_k) = & \tilde{f}(\Sigma_k) + \left\langle \nabla \tilde{f}(\Sigma_k), \Sigma - \Sigma_k \right\rangle \\ & + \frac{\phi}{2} \|\Sigma - \Sigma_k\|_F^2 + \lambda_{k+1} \|\Sigma\|_*, \end{aligned} \quad (4)$$

where ϕ is an appropriately chosen quadratic coefficient. The next iterate Σ_{k+1} is then obtained by solving the optimization problem:

$$\Sigma_{k+1} = \arg \min_{\Sigma} \tilde{g}(\Sigma | \Sigma_k)$$

Assuming that $\hat{\Sigma}$ is an exact minimizer, it satisfies the optimality condition:

$$\left\langle \hat{\Sigma} - \Sigma, \nabla \tilde{f}(\hat{\Sigma}) + \lambda \hat{\mathbf{W}} \right\rangle \leq 0,$$

where $\hat{\mathbf{W}} \in \partial \|\hat{\Sigma}\|_*$ is the subgradient of the nuclear norm. However, obtaining the exact solution $\hat{\Sigma}$ can be computationally expensive and infeasible in practical cases. To address this, we define an approximate solution to (4).

Definition 2. An approximate solution $\tilde{\Sigma}$ to (4) is said to be ε -optimal if its sub-optimal measure $\omega(\tilde{\Sigma})$ satisfies $\omega(\tilde{\Sigma}) \leq \varepsilon$, where

$$\begin{aligned} \omega(\tilde{\Sigma}) = & \min_{\tilde{\mathbf{W}} \in \partial \|\tilde{\Sigma}\|_*} \max_{\Sigma'} \left\{ \frac{\left\langle \tilde{\Sigma} - \Sigma', \nabla \tilde{f}(\tilde{\Sigma}) + \lambda \tilde{\mathbf{W}} \right\rangle}{\|\tilde{\Sigma} - \Sigma'\|_*} \right\} \\ = & \min_{\tilde{\mathbf{W}} \in \partial \|\tilde{\Sigma}\|_*} \left\{ \left\| \nabla \tilde{f}(\tilde{\Sigma}) + \lambda \tilde{\mathbf{W}} \right\|_2 \right\}, \end{aligned}$$

and the second equality follows from the duality between the nuclear norm $\|\cdot\|_*$ and the spectral norm $\|\cdot\|_2$.

The details of proximal gradient method with backtracking line search are provided in Algorithm 1.²

²Note that the positive definite check of Σ_k in Algorithm 1 can be accomplished using a Cholesky decomposition Golub and Van Loan [2013].

Algorithm 1: The Proximal Gradient Algorithm

Input: $\lambda, \varepsilon, \{y_i, \mathbf{a}_i\}_{i=1}^m$

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1 Initialize  $\eta \in (0, 1), \Sigma_0 = \mathbf{0}, \phi_0 = \phi > 0, \gamma > 1,$ 
    $\lambda_0 = \lambda, k = 1;$ 
2 repeat
3    $\lambda_k = \eta \lambda_{k-1}, \varepsilon_k = \lambda_{k-1}/4;$ 
4   repeat
5      $\Sigma_k = \arg \min_{\Sigma \succeq 0} \tilde{g}(\Sigma | \Sigma_{k-1});$ 
6     if  $\Sigma_k \not\succeq \mathbf{0}$  or  $g(\Sigma_k | \Sigma_{k-1}) < h(\Sigma_k)$  then
7        $\phi = \gamma \phi;$ 
8     end
9   until  $\Sigma_k \succeq \mathbf{0}$  and  $g(\Sigma_k | \Sigma_{k-1}) \geq h(\Sigma_k);$ 
10   $\phi = \max\{\phi_0, \gamma^{-1} \phi_t\};$ 
11   $k = k + 1;$ 
12 until  $\omega(\Sigma_k) \leq \varepsilon;$ 
Output:  $\Sigma_K$ 

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4 MAIN THEORIES

In this section, we first introduce some key technical concepts and assumptions underpinning our approach. We then present the statistical convergence rate for our proposed covariance estimator. Proof of the results is provided in the supplementary material.

4.1 BACKGROUNDS AND ASSUMPTIONS

Consider a symmetric positive semidefinite matrix $\Sigma^* \in \mathbb{R}^{d \times d}$ whose non-zero eigenvalues are denoted as the vector $\sigma^* \in \mathbb{R}^r$. The eigen-decomposition of Σ^* is given by $\Sigma^* = U^* \mathbf{\Gamma}^* U^{*\top}$, where $U^* \in \mathbb{R}^{d \times r}$ consist of orthonormal eigenvectors corresponding to the nonzero eigenvalues in σ^* , and $\mathbf{\Gamma}^* = \text{diag}(\sigma^*) \in \mathbb{R}^{r \times r}$ is a diagonal matrix containing these eigenvalues. Based on this decomposition, we define two complementary subspaces $\mathcal{S}$ and $\mathcal{S}^\perp$ as follows:

$$\mathcal{T}(U^*) := \{\Delta \mid \text{row}(\Delta) \subseteq U^* \text{ and } \text{col}(\Delta) \subseteq U^*\},$$

$$\mathcal{T}^\perp(U^*) := \{\Delta \mid \text{row}(\Delta) \perp U^* \text{ and } \text{col}(\Delta) \perp U^*\}.$$

We also define two projection operators, $\mathcal{P}(\cdot)$ and $\mathcal{P}^\perp(\cdot)$, which project matrices onto the subspaces $\mathcal{T}$ and $\mathcal{T}^\perp$, respectively. For any matrix $\Delta \in \mathbb{R}^{d \times d}$, these projections are given by:

$$\mathcal{P}(\Delta) = U^* U^{*\top} \Delta U^* U^{*\top},$$

$$\mathcal{P}^\perp(\Delta) = (I - U^* U^{*\top}) \Delta (I - U^* U^{*\top}),$$

where I denotes the identity matrix of appropriate dimension. Next, we define the subset $\mathcal{B}$, which represents a restricted set of directions:

$$\mathcal{B} = \{\Delta \in \mathbb{R}^{d \times d} \mid \|\mathcal{P}^\perp(\Delta)\|_* \leq 5 \|\mathcal{P}(\Delta)\|_*\}.$$

Assumption 3 (Restricted Strong Convexity). *For the empirical loss function $f(\cdot)$, there exists some $\rho^- > 0$ such that, for all $\Delta \in \mathcal{B}$,*

$$f(\Sigma + \Delta) \geq f(\Sigma) + \langle \nabla f(\Sigma), \Delta \rangle + \frac{\rho^-}{2} \|\Delta\|_F^2.$$

Assumption 4 (Restricted Strong Smoothness). *For the empirical loss function $f(\cdot)$, there exists some $\infty > \rho^+ \geq \rho^- > 0$ such that, for all $\Delta \in \mathcal{B}$,*

$$f(\Sigma + \Delta) \leq f(\Sigma) + \langle \nabla f(\Sigma), \Delta \rangle + \frac{\rho^+}{2} \|\Delta\|_F^2.$$

Assumption 5. *We assume that the sensing vectors $\mathbf{a}_i$'s ($1 \leq i \leq m$) are independently and identically distributed (i.i.d.) sub-Gaussian random variables with zero mean and identity covariance.*

Assumptions 3 and 4 are considered relatively mild within the nonconvex analysis Strongin and Sergeyev [2013], Gui et al. [2016]. It is well-established that, under the conditions in Assumption 5, the properties of restricted strong convexity and restricted strong smoothness hold with high probability Negahban and Wainwright [2011], Negahban et al. [2012], Loh and Wainwright [2015]. Specifically, under Assumption 5, the vectorized form of $\mathbf{a}_i \mathbf{a}_i^\top$, denoted as $\text{vec}(\mathbf{a}_i \mathbf{a}_i^\top)$, follows a sub-exponential distribution with mean zero and variance proxy Θ . Furthermore, Assumptions 3 and 4 hold with probability greater than $1 - c_1 \exp(-c_2 d)$, provided that the sample size satisfies the lower bound $m = \mathcal{O}(R_q^{1-q/2} d)$ for some constants $c_1, c_2 > 0$, $q \in [0, 1]$ and a positive radius R_q such that $R_q \geq \sum_{i=1}^d |\sigma_i(\Sigma)|^q$ Negahban and Wainwright [2011].

4.2 STATISTICAL ANALYSIS

We now present our main result, which establishes a deterministic error bound for the proposed nonconvex estimator.

Theorem 6. *Suppose that Assumptions 1, 3, and 4 hold. If $\lambda \geq 2 \|\mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma))\|_2 / m$, where $\mathcal{A}^*(\cdot)$ is the conjugate operator of $\mathcal{A}(\cdot)$, and $\rho^- > \zeta^-$, then the optimal solution $\hat{\Sigma}$ of 2 satisfies:*

$$\|\hat{\Sigma} - \Sigma^*\|_F \leq \frac{\vartheta \sqrt{|\mathcal{S}_1|}}{\rho^- - \zeta^-} + \frac{3\lambda \sqrt{|\mathcal{S}_2|}}{\rho^- - \zeta^-},$$

where $|\mathcal{S}_1|$ and $|\mathcal{S}_2|$ denote the cardinality of the sets $\mathcal{S}_1 = \{i \mid \sigma_i^* \geq \alpha\}$ and $\mathcal{S}_2 = \{i \mid \alpha > \sigma_i^* > 0\}$, respectively. The constant $\vartheta = \|\mathcal{P}_{\mathcal{S}_1}(\nabla f(\Sigma^*))\|_2$, and $\mathcal{P}_{\mathcal{S}_1}$ is the projection onto the subspace of $\mathcal{T}$ associated with $\mathcal{S}_1$.

Remark 7. *The oracle rate refers to the statistical convergence rate of the oracle estimator, which knows the true rank subspace $\mathcal{T}(\mathbf{U}^*)$ in advance. The oracle estimator $\hat{\Sigma}^O$ is defined as*

$$\hat{\Sigma}^O = \arg \min_{\Sigma: \Sigma \in \mathcal{T}(\mathbf{U}^*)} f(\Sigma).$$

The error bound derived in Theorem 6 consists of two terms, each corresponding to the larger and smaller eigenvalues of the true covariance matrix Σ^* . By selecting an appropriate value for $\zeta^- = \rho^-/2$, the proposed estimator achieves the following convergence rate:

$$\begin{aligned} \|\hat{\Sigma} - \Sigma^*\|_F &\leq \frac{2\vartheta \sqrt{|\mathcal{S}_1|}}{\rho^-} + \frac{6\lambda \sqrt{|\mathcal{S}_2|}}{\rho^-} \\ &< \frac{2\lambda \sqrt{|\mathcal{S}_1|}}{\rho^-} + \frac{6\lambda \sqrt{|\mathcal{S}_2|}}{\rho^-}, \end{aligned}$$

where the second equality follows from $\vartheta = \|\mathcal{P}_{\mathcal{S}_1}(\nabla f(\Sigma^*))\|_2 < \|\nabla f(\Sigma^*)\|_2 \leq \lambda$. This rate outperforms that of the nuclear norm-based estimator, which achieves an error bound of $\|\hat{\Sigma} - \Sigma^*\|_F \leq \frac{32\lambda \sqrt{|\mathcal{S}|}}{\rho^-}$ Negahban and Wainwright [2011]. Under some mild conditions, this estimator achieves the oracle statistical convergence rate, as detailed below.

Theorem 8 (Oracle Property). *Suppose that Assumptions 1, 3, and 4 hold. If $\rho^- > \zeta^-$, λ is selected such that*

$$\lambda \geq \frac{2 \|\mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma))\|_2}{m} + \frac{2\sqrt{r}\rho^+ \|\mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma))\|_2}{m\rho^-}$$

and σ^* satisfies

$$\min_{i \in \mathcal{S}} |\sigma_i^*| \geq \alpha + \frac{2\sqrt{r} \|\mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma))\|_2}{m\rho^-},$$

then the optimal solution $\hat{\Sigma}$ of 2 satisfies:

$$\|\hat{\Sigma} - \Sigma^*\|_F \leq \frac{2 \|\mathcal{P}(\nabla f(\Sigma^*))\|_2 \sqrt{r}}{\rho^-}.$$

As established in Theorem 8, by appropriately selecting the regularization parameter λ and ensuring that the smallest non-zero eigenvalue is sufficiently large, the proposed estimator in (2) can closely approximate the oracle estimator. This holds even in the absence of direct knowledge of the subspace $\mathcal{T}$. Consequently, the estimator in (2) is able to recover the rank of the true covariance matrix Σ^* . This shows that the proposed estimator and algorithm exhibit the oracle property, enabling near-optimal recovery of the covariance matrix. In practical terms, this implies that the covariance matrix can be estimated nearly as accurately as those obtained from full data sampling, even when the number of measurements is significantly reduced.

Next, we establish the explicit statistical convergence rate under the sub-Gaussian design.

Corollary 9. *Let $\mathbf{x}$ be a sub-Gaussian random vector with zero mean and covariance Σ^* , and let $\{\mathbf{x}_i\}_{i=1}^n$ be a collection of n i.i.d. samples drawn from $\mathbf{x}$. Suppose that Assumptions 1, 3, 4 and 5 hold. If $\lambda \geq 2C_1 \delta \varpi \sqrt{d/m}$, where $\varpi =$*

³ $\mathcal{S} = \text{supp}(\sigma^*)$ denote the set of indices corresponding to the nonzero entries of σ^*

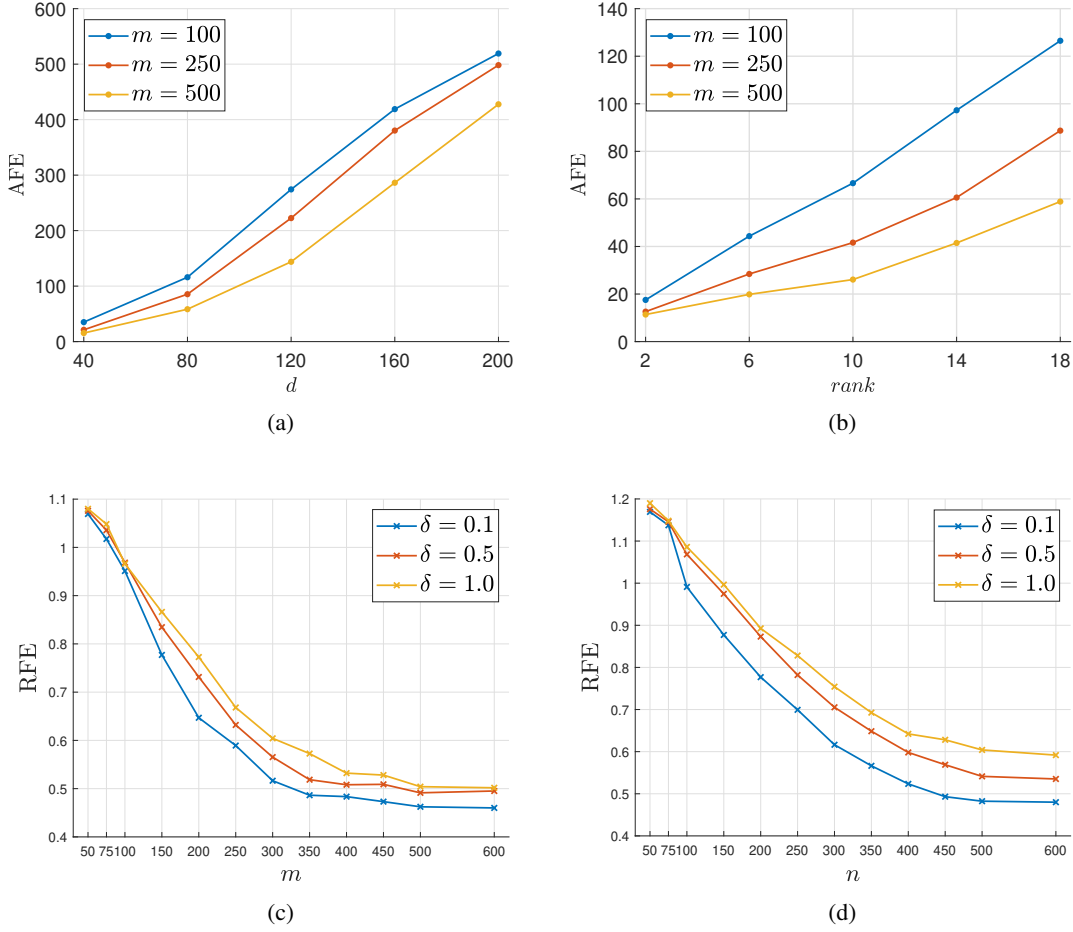


Figure 1: Absolute Frobenius norm error of the proposed estimator changing d and rank, and relative Frobenius norm error changing the noise variance proxy parameter δ .

$\sqrt{\sup_{\|\mathbf{u}\|_2=1, \|\mathbf{v}\|_2=1} \mathbf{u}^\top \mathbf{a}_i \mathbf{a}_i^\top \mathbf{v}}$ and $\rho^- = C_2 \gamma_{\min}(\boldsymbol{\Theta}) > \zeta^-$, with $\gamma_{\min}(\boldsymbol{\Theta})$ denoting the minimal eigenvalue of $\boldsymbol{\Theta}$, then with probability at least $1 - C_3 \exp(-C_4 d)$, the following inequality hold:

$$\left\| \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\|_{\text{F}} \leq \frac{\delta \varpi}{\sqrt{mn} \gamma_{\min}(\boldsymbol{\Theta})} \left[C_5 |\mathcal{S}_1| + C_6 \sqrt{|\mathcal{S}_2| d} \right].$$

Further, if $\boldsymbol{\sigma}^*$ satisfies that

$$\min_{i \in \mathcal{S}} |\sigma_i^*| \geq \nu + \frac{C_7 \delta \varpi \sqrt{|\mathcal{S}| d / m}}{\gamma_{\min}(\boldsymbol{\Theta})}.$$

And if $\lambda \geq C_8 \delta \varpi \sqrt{d/m} \left(1 + \frac{\sqrt{|\mathcal{S}|} \gamma_{\max}(\boldsymbol{\Theta})}{\gamma_{\min}(\boldsymbol{\Theta})} \right)$, with $\gamma_{\max}(\boldsymbol{\Theta})$ denoting the maximal eigenvalue of $\boldsymbol{\Theta}$, then with probability at least $1 - C_3 \exp(-C_4 d)$, the following bound holds:

$$\left\| \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\|_{\text{F}} \leq \frac{C_9 |\mathcal{S}| \delta \varpi}{\sqrt{mn} \gamma_{\min}(\boldsymbol{\Theta})},$$

where $C_1, C_2, \dots, C_9$ are universal constants.

Corollary 9 is a direct consequence of Theorem 8, which implies that under weak assumptions, we can achieve the oracle rate $\sqrt{\frac{|\mathcal{S}|}{mn}}$.

5 NUMERICAL EXPERIMENTS

In this section, we conduct a series of numerical experiments to evaluate the performance of the proposed estimator and algorithm. All experiments are implemented in MATLAB 2023b and executed on an Intel Core i7 – 10700 processor (2.90 GHz, 8 cores) with 16 GB of RAM. The results presented are averages obtained over 100 Monte Carlo simulations.

Measurement criteria and hyperparameter setup. To evaluate the performance of the proposed estimator, we use several evaluation metrics based on the Frobenius norm. Let $\hat{\boldsymbol{\Sigma}}$ denote the generic covariance matrix obtained by various algorithms and estimators. We define the absolute

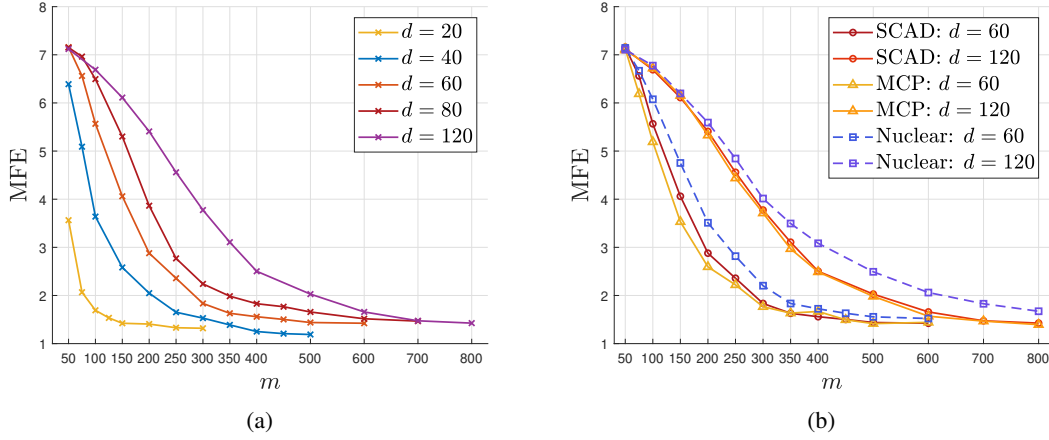


Figure 2: Mean Squared Frobenius norm error of the proposed estimator and the comparison between convex and nonconvex methods.

Table 1: The estimation error of nonconvex and convex methods $\pm$ standard variance under 100 Monte Carlo Simulations.

		$d = 40$				$d = 80$			
		$m = 100$	$m = 200$	$m = 300$	$m = 500$	$m = 100$	$m = 300$	$m = 500$	$m = 600$
SCAD	AFE	76.2997 $\pm$ 13.0608	57.2471 $\pm$ 14.4645	49.4734 $\pm$ 15.8468	43.6211 $\pm$ 18.3060	203.8819 $\pm$ 14.3887	119.7443 $\pm$ 31.1828	102.9800 $\pm$ 36.4549	98.5062 $\pm$ 46.9165
	RFE	0.7281 $\pm$ 0.1261	0.5541 $\pm$ 0.1024	0.4796 $\pm$ 0.1516	0.4087 $\pm$ 0.1627	0.9846 $\pm$ 0.0275	0.5686 $\pm$ 0.1460	0.4883 $\pm$ 0.1725	0.46850 $\pm$ 0.2120
	MFE	3.6385 $\pm$ 1.1104	2.0483 $\pm$ 1.1008	1.5298 $\pm$ 1.2044	1.1893 $\pm$ 1.4276	6.4950 $\pm$ 0.7281	2.2404 $\pm$ 1.2968	1.6570 $\pm$ 1.4817	1.5162 $\pm$ 1.2715
MCP	AFE	72.3491 $\pm$ 16.2992	52.5658 $\pm$ 18.5107	43.5744 $\pm$ 16.5545	40.0282 $\pm$ 18.4621	200.6326 $\pm$ 13.0460	120.2025 $\pm$ 33.7358	101.3771 $\pm$ 37.7358	96.7045 $\pm$ 40.9204
	RFE	0.6823 $\pm$ 0.1459	0.5050 $\pm$ 0.1631	0.4164 $\pm$ 0.1694	0.4730 $\pm$ 0.1763	0.9483 $\pm$ 0.0377	0.5705 $\pm$ 0.1579	0.4851 $\pm$ 0.1927	0.4634 $\pm$ 0.2031
	MFE	3.2715 $\pm$ 1.2436	1.7271 $\pm$ 1.3924	1.1867 $\pm$ 1.4667	1.0001 $\pm$ 1.7633	6.2896 $\pm$ 0.8240	2.2576 $\pm$ 1.3099	1.5992 $\pm$ 1.8947	1.4612 $\pm$ 1.4458
Nuclear	AFE	79.6284 $\pm$ 12.5129	58.1602 $\pm$ 14.0585	50.1646 $\pm$ 19.4004	47.1060 $\pm$ 19.2026	209.4597 $\pm$ 12.3611	122.0572 $\pm$ 36.6691	103.4399 $\pm$ 47.9068	106.7587 $\pm$ 65.1436
	RFE	0.7729 $\pm$ 0.1113	0.5656 $\pm$ 0.1397	0.4824 $\pm$ 0.1879	0.4458 $\pm$ 0.2601	0.9966 $\pm$ 0.0542	0.6408 $\pm$ 0.1837	0.4465 $\pm$ 0.2279	0.5132 $\pm$ 0.3205
	MFE	3.9629 $\pm$ 1.2688	2.1141 $\pm$ 1.0384	1.5728 $\pm$ 1.5795	1.3869 $\pm$ 2.1682	7.2163 $\pm$ 0.7761	2.4620 $\pm$ 1.3350	1.8642 $\pm$ 2.0745	1.78080 $\pm$ 3.3428
		$d = 120$				$d = 160$			
		$m = 100$	$m = 300$	$m = 500$	$m = 700$	$m = 100$	$m = 300$	$m = 500$	$m = 800$
SCAD	AFE	290.3662 $\pm$ 13.0428	233.1275 $\pm$ 34.3399	160.9030 $\pm$ 39.5538	145.6902 $\pm$ 44.8508	416.1755 $\pm$ 13.1164	371.2838 $\pm$ 20.1159	270.8453 $\pm$ 40.8805	183.1849 $\pm$ 68.6043
	RFE	0.9958 $\pm$ 0.0215	0.7469 $\pm$ 0.1097	0.54620 $\pm$ 0.1885	0.4508 $\pm$ 0.1429	1.0001 $\pm$ 0.0122	0.8987 $\pm$ 0.0435	0.6989 $\pm$ 0.0990	0.4396 $\pm$ 0.1666
	MFE	6.6894 $\pm$ 0.5615	3.7742 $\pm$ 1.0867	2.0283 $\pm$ 1.7619	1.4740 $\pm$ 0.9685	6.7657 $\pm$ 0.4275	5.4722 $\pm$ 0.5948	3.0043 $\pm$ 0.8789	1.3108 $\pm$ 1.2793
MCP	AFE	291.0940 $\pm$ 11.3085	231.1940 $\pm$ 34.4239	168.9255 $\pm$ 45.8193	145.3658 $\pm$ 52.7231	417.0620 $\pm$ 13.0938	379.877326.4551	273.8305 $\pm$ 41.9579	184.9737 $\pm$ 72.2468
	RFE	0.9915 $\pm$ 0.0234	0.7325 $\pm$ 0.1089	0.5424 $\pm$ 0.1502	0.4648 $\pm$ 0.2616	0.9966 $\pm$ 0.0121	0.9033 $\pm$ 0.0543	0.7293 $\pm$ 0.1200	0.4698 $\pm$ 0.1680
	MFE	6.7208 $\pm$ 0.4928	3.7119 $\pm$ 1.0580	1.9817 $\pm$ 1.0836	1.4674 $\pm$ 1.4120	6.7946 $\pm$ 0.4241	5.6370 $\pm$ 0.7717	3.0960 $\pm$ 0.9044	1.4850 $\pm$ 1.3038
Nuclear	AFE	292.3386 $\pm$ 8.9425	240.4273 $\pm$ 35.7671	189.4290 $\pm$ 50.5312	162.2422 $\pm$ 64.2535	416.5137 $\pm$ 15.3655	374.2606 $\pm$ 24.9383	282.5598 $\pm$ 58.3104	192.6122 $\pm$ 94.5904
	RFE	1.0010 $\pm$ 0.0174	0.7728 $\pm$ 0.1072	0.6043 $\pm$ 0.2912	0.5190 $\pm$ 0.2086	1.0014 $\pm$ 0.0130	0.9004 $\pm$ 0.0548	0.6810 $\pm$ 0.1414	0.46160 $\pm$ 0.1530
	MFE	6.7747 $\pm$ 0.3910	4.0143 $\pm$ 1.2636	2.4919 $\pm$ 3.6096	1.8280 $\pm$ 1.6649	6.7767 $\pm$ 0.4999	5.4715 $\pm$ 0.7326	3.1188 $\pm$ 1.3402	1.4492 $\pm$ 1.1249

Frobenius error (AFE) as $\|\hat{\Sigma} - \Sigma^*\|_F$, where Σ^* represent the true covariance matrix. Additionally, the mean squared Frobenius error (MFE) is given by $\frac{\|\hat{\Sigma} - \Sigma^*\|_F^2}{d^2}$, where d is the dimension of the covariance matrix. Similarly, the relative Frobenius error (RFE) is defined as $\frac{\|\hat{\Sigma} - \Sigma^*\|_F}{\|\Sigma^*\|_F}$. For all methods, the tuning parameter λ is selected using five-fold cross-validation. Unless otherwise specified, we employ the SCAD penalty with a parameter $b = 3.7$, and set the variance proxy $\delta = 0.1$.

5.1 SYNTHETIC DATA

In our experiments, synthetic datasets are generated to evaluate the performance under various conditions. The true covariance matrix, denoted as Σ^* , is constructed using a low-rank model of the form $\Sigma^* = LL^\top$, where L is a $d \times r$ matrix whose components are independently sampled

from a standard Gaussian distribution. The sensing vector $\{a_i\}_{i=1}^m$ are drawn independently from a Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. In addition, the noise $\{\eta_i\}_{i=1}^m$ are sampled from a sub-Gaussian distribution with mean $\mathbf{0}$ and a variance proxy of δ^2 . The experiments explore a range of scenarios where the dimensions d , ranks r , the measurement sizes m , and the sample size n .

In Figure 1a, we show how the AFE evolves as the dimension d of the covariance matrix increases, with the rank fixed at $r = 10$. The curves represent different sample sizes, specifically $m = \{100, 300, 500\}$, and clearly illustrate that as the dimension d grows, the estimation error increases. This trend suggests that a larger sample size is necessary to achieve convergence as the problem dimension expands.

Figure 1b presents the results for a fixed matrix dimension $d = 60$, where we investigate the impact of varying the rank of the true covariance matrix. As the rank increases,

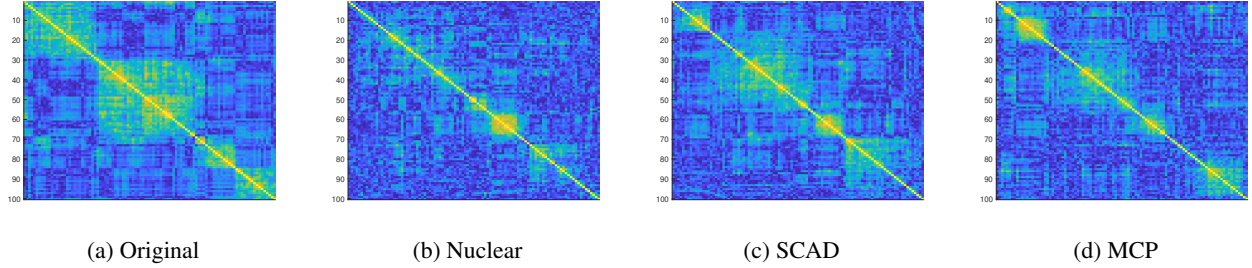


Figure 3: The heatmaps on the absolute values of the estimated correlations among SRBCT gene values.

Table 2: Performance comparison on the SRBCT gene expression dataset under 100 Monte Carlo Simulations.

Dataset	Metric	SCAD	MCP	Nuclear
Gene-Data	MFE	12.5116 ± 1.7491	13.0716 ± 1.7456	15.7959 ± 1.9235
	RFE	0.3324 ± 0.0927	0.3954 ± 0.0884	0.5949 ± 0.1027
	RMSE	0.1828 ± 0.0196	0.2063 ± 0.0178	0.7019 ± 0.0279

we observe a corresponding rise in the estimation error, indicating that higher-rank models pose more challenging estimation tasks.

Figure 1c illustrates the effect of measurement size and noise intensity on the estimation performance, with dimension d fixed at 60 and sample size n fixed at 50. As the noise variance proxy δ increases, the estimation errors also rise, indicating the estimator’s sensitivity to the noise component. Furthermore, we observe that larger measurement sizes lead to improved estimation accuracy, underscoring the advantage of increasing the number of measurements.

Figure 1d shows the influence of sample size and noise intensity on the estimation performance, with dimension d fixed at 60 and measurement size m fixed at 300. Again, increasing the noise level (via the variance proxy δ) results in higher estimation errors. Additionally, larger sample sizes yield more accurate estimations, highlighting the benefits of increasing the number of samples for improved performance.

Figures 2a and 2b present results from experiments using synthetic data, where the number of measurements is varied while the matrix rank is kept fixed at 10 and the sample size is kept fixed at $n = 50$. Figure 2a shows the mean squared Frobenius norm error of our proposed nonconvex estimator. As the matrix dimension increases, the estimation error rapidly converges to a constant value. Figure 2b compares the performance of the nonconvex method with the convex method based on the Nuclear Norm. The results indicate that the nonconvex method achieves a faster rate of error reduction as the sample size increases, demonstrating its superior performance.

Table 1 presents the performance of different penalty func-

tions and covariance matrix dimensions. The table displays the statistical results for SCAD, MCP, and Nuclear Norm approaches, under various estimation criteria and standard deviations. In each experiment, the rank of the covariance matrix is fixed at $r = 10$, the sample size is fixed at $n = 50$, and the variance proxy is set to $\delta = 0.1$. The dimensions of the estimated matrix vary within the set $d = 40, 80, 120, 160$.

5.2 REAL-WORLD DATASETS

We present experimental results on the SRBCT gene expression data Khan et al. [2001], which includes expression levels of 2038 genes across 83 samples. For this dataset, we compute the F-statistic for all gene expression values and select the top 100 genes for further estimation. The data is transformed into a covariance matrix, and the sensing vector $\mathbf{a}_i$ is generated accordingly. We set the measurement number to $m = 500$ and choose a variance proxy of $\delta = 0.1$. To evaluate the performance of our proposed estimator, we compare three different approaches: SCAD, MCP, and Nuclear, with the parameter $b = 3.7$. The Root Mean Square Error (RMSE) is defined as $\sqrt{\frac{1}{m} \sum_{i=1}^m \left\| y_i - \langle \mathbf{a}_i \mathbf{a}_i^\top, \hat{\Sigma} \rangle \right\|_2^2}$ and the MFE and RFE for each method are summarized in Table 2. Additionally, we visualize of the correlations among the selected 100 genes. These results are shown in the Figure 3, where the heatmaps represent the absolute values of the estimated correlations among these genes.

6 CONCLUSIONS

In this paper, we address the problem of recovering a low-rank covariance matrix from a quadratic measurement

model, which effectively mitigates the challenges posed by limited measurements, as well as the constraints on storage and computational resources common in high-dimensional settings. We introduce a novel nonconvex penalization approach and employ a proximal gradient homotopy algorithm to solve the resulting estimator. Our approach demonstrates superior statistical convergence rates compared to traditional nuclear norm penalization methods, and under mild assumptions, it achieves the oracle property. These theoretical advancements are supported by comprehensive empirical evaluations.

Author Contributions

Briefly list author contributions. This is a nice way of making clear who did what and to give proper credit. This section is optional.

H. Q. Bovik conceived the idea and wrote the paper. Coauthor One created the code. Coauthor Two created the figures.

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Low-rank Covariance Matrix Estimation from Quadratic Measurements (Supplementary Material)

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A APPENDICES

A.1 TECHNICAL LEMMATA

Lemma 10. *Under Assumption 3, and assuming that $\Sigma_1 - \Sigma_2 \in \mathcal{B}$, we establish*

$$\tilde{f}(\Sigma_2) \geq \tilde{f}(\Sigma_1) + \langle \nabla \tilde{f}(\Sigma_1), \Sigma_2 - \Sigma_1 \rangle + \frac{\rho^- - \zeta^-}{2} \|\Sigma_2 - \Sigma_1\|_F^2.$$

Proof. Considering the eigen-decompositions of covariance matrices Σ_1 and Σ_2 given by

$$\Sigma_1 = U_1 \Gamma_1 U_1^\top, \Sigma_2 = U_2 \Gamma_2 U_2^\top,$$

where $\Gamma_1 = \text{diag}(\sigma_1)$ and $\Gamma_2 = \text{diag}(\sigma_2)$ are diagonal matrices containing eigenvalues σ_1 and σ_2 of Σ_1 and Σ_2 , respectively.

Under (b) of Assumption 1, for each component $i = 1, 2, \dots, d$ in σ_1 and σ_2 , we have

$$q'_\lambda((\sigma_1)_i) - q'_\lambda((\sigma_2)_i) \geq -\zeta^-((\sigma_1)_i - (\sigma_2)_i),$$

which leads to

$$\langle (-\nabla Q_\lambda(\Sigma_1)) - (-\nabla Q_\lambda(\Sigma_2)), \Sigma_1 - \Sigma_2 \rangle \leq \zeta^- \|\Sigma_1 - \Sigma_2\|_F^2.$$

This inequality illustrates the strong smoothness of $-Q(\cdot)$ in the region $\mathcal{B}$. As a consequence, we establish

$$Q_\lambda(\Sigma_1) + \langle \nabla Q_\lambda(\Sigma_1), \Sigma_2 - \Sigma_1 \rangle - \frac{\zeta^-}{2} \|\Sigma_1 - \Sigma_2\|_F^2 \leq Q_\lambda(\Sigma_2). \quad (5)$$

Additionally, by Assumption 3, we obtain

$$f(\Sigma_2) \geq f(\Sigma_1) + \langle \nabla f(\Sigma_1), \Sigma_2 - \Sigma_1 \rangle + \frac{\rho^-}{2} \|\Sigma_2 - \Sigma_1\|_F^2. \quad (6)$$

Given that $\tilde{f}(\Sigma)$ is the sum of $f(\Sigma)$ and $Q_\lambda(\Sigma)$, we can combine (5) and (6) to achieve

$$\tilde{f}(\Sigma_2) \geq \tilde{f}(\Sigma_1) + \langle \nabla \tilde{f}(\Sigma_1), \Sigma_2 - \Sigma_1 \rangle + \frac{\rho^- - \zeta^-}{2} \|\Sigma_2 - \Sigma_1\|_F^2.$$

This result demonstrates the restricted strong convexity of the objective function $\tilde{f}(\Sigma)$. □

Lemma 11. Suppose Assumption 3 holds. If $\rho^- > \zeta^-$, and $\lambda \geq \frac{2}{m} \|\mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))\|_2$, we have

$$\left\| \mathcal{P}^\perp \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\|_* \leq 5 \left\| \mathcal{P} \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\|_*.$$

Proof. Given Lemma 10, we obtain

$$\underbrace{\left(\tilde{f}(\hat{\boldsymbol{\Sigma}}) + \lambda \|\hat{\boldsymbol{\Sigma}}\|_* \right) - \left(\tilde{f}(\boldsymbol{\Sigma}^*) + \lambda \|\boldsymbol{\Sigma}^*\|_* \right)}_{\text{I}} \geq \underbrace{\left\langle \nabla \tilde{f}(\boldsymbol{\Sigma}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle}_{\text{II}} + \underbrace{\lambda \|\hat{\boldsymbol{\Sigma}}\|_* - \lambda \|\boldsymbol{\Sigma}^*\|_*}_{\text{III}}. \quad (7)$$

In the following, we analyze the term II. Applying Hölder's inequality, we have

$$\begin{aligned} & \left\langle \nabla \tilde{f}(\boldsymbol{\Sigma}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \\ &= \left\langle \nabla \tilde{f}(\boldsymbol{\Sigma}^*), \mathcal{P} \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\rangle + \left\langle \nabla \tilde{f}(\boldsymbol{\Sigma}^*), \mathcal{P}^\perp \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\rangle \\ &\geq - \underbrace{\left\| \mathcal{P} \left(\nabla \tilde{f}(\boldsymbol{\Sigma}^*) \right) \right\|_2}_{\text{II}_1} \cdot \left\| \mathcal{P} \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\|_* - \underbrace{\left\| \mathcal{P}^\perp \left(\nabla \tilde{f}(\boldsymbol{\Sigma}^*) \right) \right\|_2}_{\text{II}_2} \cdot \left\| \mathcal{P}^\perp \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\|_*. \end{aligned} \quad (8)$$

It remains to bound the terms I, II₁, II₂ and III, respectively.

For the term I, since $\hat{\boldsymbol{\Sigma}}$ represents the optimal solution to the problem (2). This leads to

$$\left(\tilde{f}(\hat{\boldsymbol{\Sigma}}) + \lambda \|\hat{\boldsymbol{\Sigma}}\|_* \right) - \left(\tilde{f}(\boldsymbol{\Sigma}^*) + \lambda \|\boldsymbol{\Sigma}^*\|_* \right) \leq 0. \quad (9)$$

For the term II₁, considering the choice of λ , we have

$$\|\nabla f(\boldsymbol{\Sigma}^*)\|_2 = \left\| \frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*)) \right\|_2 \leq \frac{\lambda}{2}.$$

Based on (a) in Assumption 1 and $\lambda \geq \frac{2}{m} \|\mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))\|_2$, we have $\|\nabla Q_\lambda(\boldsymbol{\Sigma}^*)\|_2 \leq \lambda$, which leads to

$$\left\| \mathcal{P} \left(\nabla \tilde{f}(\boldsymbol{\Sigma}^*) \right) \right\|_2 = \left\| \mathcal{P} \left(\nabla f(\boldsymbol{\Sigma}^*) + \nabla Q_\lambda(\boldsymbol{\Sigma}^*) \right) \right\|_2 \leq \frac{3\lambda}{2}. \quad (10)$$

For term II₂, we first note from the definition of $\mathcal{P}^\perp(\cdot)$ that $\mathcal{P}^\perp(\boldsymbol{\Sigma}^*) = \mathbf{0}$. Based on (c) in Assumption 1, it follows that $Q_\lambda(\mathcal{P}^\perp(\boldsymbol{\Sigma}^*)) = 0$. This leads to

$$\left\| \mathcal{P}^\perp \left(\nabla \tilde{f}(\boldsymbol{\Sigma}^*) \right) \right\|_2 = \left\| \mathcal{P}^\perp \left(\nabla f(\boldsymbol{\Sigma}^*) \right) \right\|_2 \leq \frac{\lambda}{2}. \quad (11)$$

Then we substitute (10) and (11) into (8), we have

$$\left\langle \nabla \tilde{f}(\boldsymbol{\Sigma}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \geq -\frac{3\lambda}{2} \left\| \mathcal{P} \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\|_* - \frac{\lambda}{2} \left\| \mathcal{P}^\perp \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\|_*. \quad (12)$$

For the term III, projecting onto the complementary subspaces $\mathcal{T}$ and $\mathcal{T}^\perp$ yields the following decomposition

$$\begin{aligned} \lambda \|\hat{\boldsymbol{\Sigma}}\|_* - \lambda \|\boldsymbol{\Sigma}^*\|_* &= \lambda \|\mathcal{P}(\hat{\boldsymbol{\Sigma}})\|_* - \lambda \|\mathcal{P}(\boldsymbol{\Sigma}^*)\|_* + \lambda \|\mathcal{P}^\perp(\hat{\boldsymbol{\Sigma}})\|_* - \lambda \|\mathcal{P}^\perp(\boldsymbol{\Sigma}^*)\|_* \\ &\geq -\lambda \left\| \mathcal{P} \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\|_* + \lambda \left\| \mathcal{P}^\perp \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\|_*. \end{aligned} \quad (13)$$

Based on the inequalities in formulas (9), (12), and (13), when substituted into formula (7), we obtain

$$0 \geq -\frac{3\lambda}{2} \left\| \mathcal{P} \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\|_* - \frac{\lambda}{2} \left\| \mathcal{P}^\perp \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\|_* - \lambda \left\| \mathcal{P} \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\|_* + \lambda \left\| \mathcal{P}^\perp \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\|_*.$$

From this inequality, we have

$$\left\| \mathcal{P}^\perp \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\|_* \leq 5 \left\| \mathcal{P} \left(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right) \right\|_*,$$

which implies that $\hat{\boldsymbol{\Delta}} = \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^*$ lies in the cone $\mathcal{B}$. $\square$

A.2 PROOF OF THEOREM 6

Proof. Applying Lemma 10, we establish

$$\begin{aligned}\tilde{f}(\hat{\Sigma}) &\geq \tilde{f}(\Sigma^*) + \langle \nabla \tilde{f}(\Sigma^*), \hat{\Sigma} - \Sigma^* \rangle + \frac{\rho^- - \zeta^-}{2} \|\hat{\Sigma} - \Sigma^*\|_F^2, \\ \tilde{f}(\Sigma^*) &\geq \tilde{f}(\hat{\Sigma}) + \langle \nabla \tilde{f}(\hat{\Sigma}), \Sigma^* - \hat{\Sigma} \rangle + \frac{\rho^- - \zeta^-}{2} \|\Sigma^* - \hat{\Sigma}\|_F^2.\end{aligned}$$

By summing these two inequalities, we obtain

$$(\rho^- - \zeta^-) \|\hat{\Sigma} - \Sigma^*\|_F^2 - \langle \nabla \tilde{f}(\hat{\Sigma}) - \nabla \tilde{f}(\Sigma^*), \hat{\Sigma} - \Sigma^* \rangle \leq 0. \quad (14)$$

Considering the convexity of the nuclear norm $\|\cdot\|_*$, we deduce

$$\langle \lambda \hat{W} - \lambda W^*, \hat{\Sigma} - \Sigma^* \rangle \geq 0, \quad (15)$$

where $\hat{W} \in \partial \|\hat{\Sigma}\|_*$ and $W^* \in \partial \|\Sigma^*\|_*$.

Since $\hat{\Sigma}$ is the optimal solution to (2), it satisfies the optimality condition:

$$\langle \nabla \tilde{f}(\hat{\Sigma}) + \lambda \hat{W}, \hat{\Sigma} - \Sigma^* \rangle \leq 0. \quad (16)$$

By combining the inequalities (14), (15), and (16), we obtain

$$\begin{aligned}(\rho^- - \zeta^-) \|\hat{\Sigma} - \Sigma^*\|_F^2 &\leq \langle \nabla \tilde{f}(\hat{\Sigma}) - \nabla \tilde{f}(\Sigma^*), \hat{\Sigma} - \Sigma^* \rangle \\ &= \langle \nabla \tilde{f}(\hat{\Sigma}) + \lambda \hat{W}, \hat{\Sigma} - \Sigma^* \rangle - \langle \lambda \hat{W} - \lambda W^*, \hat{\Sigma} - \Sigma^* \rangle - \langle \nabla \tilde{f}(\Sigma^*) + \lambda W^*, \hat{\Sigma} - \Sigma^* \rangle \\ &\leq - \langle \nabla \tilde{f}(\Sigma^*) + \lambda W^*, \hat{\Sigma} - \Sigma^* \rangle.\end{aligned} \quad (17)$$

Consider the partitioning of the eigenvalues of Σ^* into three distinct sets:

- For indices $i \in \mathcal{S}_1$, the eigenvalues satisfy $\sigma_i^* \geq \alpha$.
- For indices $i \in \mathcal{S}_2$, we have $\alpha > \sigma_i^* > 0$.
- For indices $i \in \bar{\mathcal{S}}$, we have $\sigma_i^* = 0$, with $\mathcal{S} = \mathcal{S}_1 \cup \mathcal{S}_2$.

Based on this partition, we define two subspace of $\mathcal{T}$ associated with $\mathcal{S}_1$ and $\mathcal{S}_2$ as follows:

$$\mathcal{T}_{\mathcal{S}_1}(U^*) := \{ \Delta \in \mathbb{R}^{d \times d} \mid \text{row}(\Delta) \subset U_{\mathcal{S}_1}^* \text{ and } \text{col}(\Delta) \subset U_{\mathcal{S}_1}^* \},$$

$$\mathcal{T}_{\mathcal{S}_2}(U^*) := \{ \Delta \in \mathbb{R}^{d \times d} \mid \text{row}(\Delta) \subset U_{\mathcal{S}_2}^* \text{ and } \text{col}(\Delta) \subset U_{\mathcal{S}_2}^* \},$$

where $U_{\mathcal{S}_1}^*$ is the sub-matrix of U^* formed by the rows indexed by $i \in \mathcal{S}_1$, and similarly, $U_{\mathcal{S}_2}^*$ is the sub-matrix formed by the rows indexed by $i \in \mathcal{S}_2$. Corresponding to these subspaces, we define two projection operators $\mathcal{P}_{\mathcal{S}_1}(\cdot)$ and $\mathcal{P}_{\mathcal{S}_2}(\cdot)$, which project matrices onto the subspace $\mathcal{T}_{\mathcal{S}_1}$ and $\mathcal{T}_{\mathcal{S}_2}$, respectively. In the following, we decompose the RHS in (17) with respect to these three distinct components.

For $i \in \bar{\mathcal{S}}$, we proceed under (a) in Assumption 1, which ensures that $q'_\lambda(0) = 0$. This property enables us to express the gradient as follows:

$$\nabla Q_\lambda(\Sigma^*) = U^* q'_\lambda(I^*) U^{*\top},$$

where $\mathbf{\Gamma}^* \in \mathbb{R}^{r \times r}$ is the diagonal matrix with $\text{diag}(\mathbf{\Gamma}^*) = \boldsymbol{\sigma}^*$. By projecting $\nabla Q_\lambda(\boldsymbol{\Sigma}^*)$ onto $\mathcal{T}^\perp$, we obtain

$$\begin{aligned} \mathcal{P}^\perp(\nabla Q_\lambda(\boldsymbol{\Sigma}^*)) &= \left(\mathbf{I} - \mathbf{U}^* \mathbf{U}^{*\top} \right) \mathbf{U}^* q'_\lambda(\mathbf{\Gamma}^*) \mathbf{U}^{*\top} \left(\mathbf{I} - \mathbf{U}^* \mathbf{U}^{*\top} \right) \\ &= (\mathbf{U}^* - \mathbf{U}^*) q'_\lambda(\mathbf{\Gamma}^*) (\mathbf{U}^{*\top} - \mathbf{U}^{*\top}) \\ &= \mathbf{0}. \end{aligned} \quad (18)$$

Subsequently, we decompose $\mathbf{W}^* \in \partial \|\boldsymbol{\Sigma}^*\|_*$ as:

$$\mathbf{W}^* = \mathcal{P}(\mathbf{W}^*) + \mathcal{P}^\perp(\mathbf{W}^*) = \mathbf{U}^* \mathbf{U}^{*\top} - \lambda^{-1} \mathcal{P}^\perp(\nabla f(\boldsymbol{\Sigma}^*)), \quad (19)$$

which follows from the norm condition:

$$\|\lambda^{-1} \mathcal{P}^\perp(\nabla f(\boldsymbol{\Sigma}^*))\|_2 \leq \lambda^{-1} \|\nabla f(\boldsymbol{\Sigma}^*)\|_2 = \lambda^{-1} m^{-1} \|\mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}))\|_2 \leq 1.$$

Combining the results from (18) and (19), we obtain

$$\begin{aligned} \left\langle \mathcal{P}^\perp(\nabla \tilde{f}(\boldsymbol{\Sigma}^*) + \lambda \mathbf{W}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle &= \left\langle \mathcal{P}^\perp(\nabla f(\boldsymbol{\Sigma}^*)) + \lambda \mathcal{P}^\perp(\mathbf{W}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \\ &= \left\langle \mathbf{0}, \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \\ &= 0. \end{aligned} \quad (20)$$

For $i \in \mathcal{S}_1$, under (d) in Assumption 1, since $\sigma_i^* \geq \alpha$, we have $q'_\lambda(\sigma_i^*) = -\lambda$. Consequently, we can project $\nabla Q_\lambda(\boldsymbol{\Sigma}^*)$ onto $\mathcal{T}_{\mathcal{S}_1}$ as follows:

$$\mathcal{P}_{\mathcal{S}_1}(\nabla Q_\lambda(\boldsymbol{\Sigma}^*)) = \mathbf{U}_{\mathcal{S}_1}^* q'_\lambda(\mathbf{\Gamma}_{\mathcal{S}_1}^*) \mathbf{U}_{\mathcal{S}_1}^{*\top} = -\lambda \mathbf{U}_{\mathcal{S}_1}^* \mathbf{U}_{\mathcal{S}_1}^{*\top},$$

where $\mathbf{\Gamma}_{\mathcal{S}_1}^* \in \mathbb{R}^{r \times r}$. Then, we have

$$\begin{aligned} &\left\langle \mathcal{P}_{\mathcal{S}_1}(\nabla \tilde{f}(\boldsymbol{\Sigma}^*) + \lambda \mathbf{W}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \\ &= \left\langle \mathcal{P}_{\mathcal{S}_1}(\nabla f(\boldsymbol{\Sigma}^*)) + \mathcal{P}_{\mathcal{S}_1}(\lambda \mathbf{W}^*) + \mathcal{P}_{\mathcal{S}_1}(\nabla Q_\lambda(\boldsymbol{\Sigma}^*)), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \\ &= \left\langle \mathcal{P}_{\mathcal{S}_1}(\nabla f(\boldsymbol{\Sigma}^*)) + \mathcal{P}_{\mathcal{S}_1}(\lambda \mathbf{U}^* \mathbf{U}^{*\top}) - \lambda \mathbf{U}_{\mathcal{S}_1}^* \mathbf{U}_{\mathcal{S}_1}^{*\top}, \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \\ &= \left\langle \mathcal{P}_{\mathcal{S}_1}(\nabla f(\boldsymbol{\Sigma}^*)), \mathcal{P}_{\mathcal{S}_1}(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^*) \right\rangle. \end{aligned} \quad (21)$$

By applying Hölder's inequality and define $\vartheta = \|\mathcal{P}_{\mathcal{S}_1}(\nabla f(\boldsymbol{\Sigma}^*))\|_2$, we update (21) to

$$\left\langle \mathcal{P}_{\mathcal{S}_1}(\nabla \tilde{f}(\boldsymbol{\Sigma}^*) + \lambda \mathbf{W}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \geq -\vartheta \left\| \mathcal{P}_{\mathcal{S}_1}(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^*) \right\|_*. \quad (22)$$

Since $\text{rank}(\mathcal{P}_{\mathcal{S}_1}(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^*)) \leq |\mathcal{S}_1|$, we further refine (22) as

$$\left\langle \mathcal{P}_{\mathcal{S}_1}(\nabla \tilde{f}(\boldsymbol{\Sigma}^*) + \lambda \mathbf{W}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \geq -\vartheta \sqrt{|\mathcal{S}_1|} \left\| \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\|_{\text{F}}.$$

For $i \in \mathcal{S}_2$, applying Hölder's inequality, we derive

$$\begin{aligned} &\left\langle \mathcal{P}_{\mathcal{S}_2}(\nabla \tilde{f}(\boldsymbol{\Sigma}^*) + \lambda \mathbf{W}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \\ &= \left\langle \mathcal{P}_{\mathcal{S}_2}(\nabla f(\boldsymbol{\Sigma}^*) + \nabla Q_\lambda(\boldsymbol{\Sigma}^*) + \lambda \mathbf{W}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \\ &= \left\langle \mathcal{P}_{\mathcal{S}_2}(\nabla f(\boldsymbol{\Sigma}^*)), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle + \left\langle \mathcal{P}_{\mathcal{S}_2}(\nabla Q_\lambda(\boldsymbol{\Sigma}^*)), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle + \left\langle \mathcal{P}_{\mathcal{S}_2}(\lambda \mathbf{W}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \\ &\geq - \left[\underbrace{\left\| \mathcal{P}_{\mathcal{S}_2}(\nabla f(\boldsymbol{\Sigma}^*)) \right\|_2}_{\text{I}} + \underbrace{\left\| \mathcal{P}_{\mathcal{S}_2}(\nabla Q_\lambda(\boldsymbol{\Sigma}^*)) \right\|_2}_{\text{II}} + \underbrace{\left\| \mathcal{P}_{\mathcal{S}_2}(\lambda \mathbf{W}^*) \right\|_2}_{\text{III}} \right] \cdot \left\| \mathcal{P}_{\mathcal{S}_2}(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^*) \right\|_2. \end{aligned} \quad (23)$$

It remains to analysis the terms I, II and III in (23) individually.

For term I, we have

$$\|\mathcal{P}_{\mathcal{S}_2}(\nabla f(\boldsymbol{\Sigma}^*))\|_2 \leq \|\nabla f(\boldsymbol{\Sigma}^*)\|_2 \leq \lambda. \quad (24)$$

For term II, we have

$$\|\mathcal{P}_{\mathcal{S}_2}(\nabla Q_\lambda(\boldsymbol{\Sigma}^*))\|_2 = \left\| \mathbf{U}_{\mathcal{S}_1}^* q'_\lambda(\boldsymbol{\Gamma}_{\mathcal{S}_1}^*) \mathbf{U}_{\mathcal{S}_2}^{*\top} \right\|_2.$$

Additionally, under (a) in Assumption 1, for any $i \in \mathcal{S}_2$, σ_i^* satisfies $\alpha > \sigma_i^* > 0$, we have $|q'_\lambda(\sigma_i^*)| \leq \lambda$. This leads to

$$\|\mathcal{P}_{\mathcal{S}_2}(\nabla Q_\lambda(\boldsymbol{\Sigma}^*))\|_2 \leq \lambda. \quad (25)$$

For term III, we have

$$\|\mathcal{P}_{\mathcal{S}_2}(\lambda \mathbf{W}^*)\|_2 \leq \left\| \mathcal{P}_{\mathcal{S}_2}(\lambda \mathbf{U}^* \mathbf{U}^{*\top}) \right\|_2 = \lambda. \quad (26)$$

By substituting (24), (25) and (26) into (23), we have

$$\left\langle \mathcal{P}_{\mathcal{S}_2}(\nabla \tilde{f}(\boldsymbol{\Sigma}^*) + \lambda \mathbf{W}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \geq -3\lambda \left\| \mathcal{P}_{\mathcal{S}_2}(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^*) \right\|_2.$$

Since $\text{rank}(\mathcal{P}_{\mathcal{S}_2}(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^*)) \leq |\mathcal{S}_2|$, we have

$$\left\langle \mathcal{P}_{\mathcal{S}_2}(\nabla \tilde{f}(\boldsymbol{\Sigma}^*) + \lambda \mathbf{W}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \geq -3\lambda \sqrt{|\mathcal{S}_2|} \left\| \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\|_F. \quad (27)$$

By substituting (20), (22) and (27) into (17), we have

$$\begin{aligned} & (\rho^- - \zeta^-) \left\| \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\|_F^2 \\ & \leq - \left\langle \mathcal{P}_{\mathcal{S}_1}(\nabla \tilde{f}(\boldsymbol{\Sigma}^*) + \lambda \mathbf{W}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle - \left\langle \mathcal{P}_{\mathcal{S}_2}(\nabla \tilde{f}(\boldsymbol{\Sigma}^*) + \lambda \mathbf{W}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \\ & \quad - \left\langle \mathcal{P}^\perp(\nabla \tilde{f}(\boldsymbol{\Sigma}^*) + \lambda \mathbf{W}^*), \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\rangle \\ & \leq \vartheta \sqrt{|\mathcal{S}_1|} \left\| \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\|_F + 3\lambda \sqrt{|\mathcal{S}_2|} \left\| \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\|_F. \end{aligned}$$

Consequently, we get

$$\left\| \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\|_F \leq \frac{\vartheta \sqrt{|\mathcal{S}_1|}}{\rho^- - \zeta^-} + \frac{3\lambda \sqrt{|\mathcal{S}_2|}}{\rho^- - \zeta^-}.$$

□

A.3 PROOF OF THEOREM 8

Proof. According to the definitions of $f(\cdot)$, we have

$$\begin{aligned} f(\hat{\boldsymbol{\Sigma}}^O) - f(\boldsymbol{\Sigma}^*) &= \frac{1}{2m} \left\| \mathbf{y} - \mathcal{A}(\hat{\boldsymbol{\Sigma}}^O) \right\|_2^2 - \frac{1}{2m} \left\| \mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*) \right\|_2^2 \\ &= \frac{1}{2m} \left\| \boldsymbol{\eta} - \mathcal{A}(\hat{\boldsymbol{\Sigma}}^O - \boldsymbol{\Sigma}^*) \right\|_2^2 - \frac{1}{2m} \left\| \boldsymbol{\eta} \right\|_2^2 \\ &= \frac{1}{2m} \left\| \mathcal{A}(\hat{\boldsymbol{\Sigma}}^O - \boldsymbol{\Sigma}^*) \right\|_2^2 - \frac{1}{m} \left\langle \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*)), \hat{\boldsymbol{\Sigma}}^O - \boldsymbol{\Sigma}^* \right\rangle \\ &= \frac{1}{2m} \left\| \mathcal{A}(\hat{\boldsymbol{\Sigma}}^O - \boldsymbol{\Sigma}^*) \right\|_2^2 + \left\langle \nabla f(\boldsymbol{\Sigma}^*), \hat{\boldsymbol{\Sigma}}^O - \boldsymbol{\Sigma}^* \right\rangle \leq 0, \end{aligned} \quad (28)$$

where the last inequality applies due to $\hat{\Sigma}^O$ is the oracle estimator.

Next, applying the assumption 3, we have

$$f(\hat{\Sigma}^O) \geq f(\Sigma^*) + \langle \nabla f(\Sigma^*), \hat{\Sigma}^O - \Sigma^* \rangle + \frac{\rho^-}{2} \|\hat{\Sigma}^O - \Sigma^*\|_F^2. \quad (29)$$

By combining inequalities (28) and (29), we obtain

$$\begin{aligned} \frac{\rho^-}{2} \|\hat{\Sigma}^O - \Sigma^*\|_F^2 &\leq f(\hat{\Sigma}^O) - f(\Sigma^*) - \langle \nabla f(\Sigma^*), \hat{\Sigma}^O - \Sigma^* \rangle \\ &= \frac{1}{2m} \|\mathcal{A}(\hat{\Sigma}^O - \Sigma^*)\|_2^2 \\ &\leq -\langle \nabla f(\Sigma^*), \hat{\Sigma}^O - \Sigma^* \rangle. \end{aligned}$$

This inequality can be further refined using Hölder's inequality as follows

$$\|\hat{\Sigma}^O - \Sigma^*\|_F^2 \leq \frac{2 \langle \mathcal{P}(\nabla f(\Sigma^*)), \hat{\Sigma}^O - \Sigma^* \rangle}{\rho^-} \leq \frac{2 \|\mathcal{P}(\nabla f(\Sigma^*))\|_2 \cdot \|\hat{\Sigma}^O - \Sigma^*\|_*}{\rho^-}. \quad (30)$$

Given that $\text{rank}(\hat{\Sigma}^O - \Sigma^*) = r$, we have $\|\hat{\Sigma}^O - \Sigma^*\|_* \leq \sqrt{r} \|\hat{\Sigma}^O - \Sigma^*\|_F$. Using this in (30) gives our desired result

$$\|\hat{\Sigma}^O - \Sigma^*\|_F^2 \leq \frac{2\sqrt{r} \|\mathcal{P}(\nabla f(\Sigma^*))\|_2 \cdot \|\hat{\Sigma}^O - \Sigma^*\|_F}{\rho^-}.$$

Thus, we have

$$\|\hat{\Sigma}^O - \Sigma^*\|_F \leq \frac{2\sqrt{r} \|\mathcal{P}(\nabla f(\Sigma^*))\|_2}{\rho^-}.$$

Next, we will prove that $\hat{\Sigma}^O = \hat{\Sigma}$ under our assumptions. From Lemma 10, the following inequalities holds

$$\begin{aligned} \tilde{f}(\hat{\Sigma}) &\geq \tilde{f}(\hat{\Sigma}^O) + \langle \nabla \tilde{f}(\hat{\Sigma}^O), \hat{\Sigma} - \hat{\Sigma}^O \rangle + \frac{\rho^- - \zeta^-}{2} \|\hat{\Sigma} - \hat{\Sigma}^O\|_F^2, \\ \tilde{f}(\hat{\Sigma}^O) &\geq \tilde{f}(\hat{\Sigma}) + \langle \nabla \tilde{f}(\hat{\Sigma}), \hat{\Sigma}^O - \hat{\Sigma} \rangle + \frac{\rho^- - \zeta^-}{2} \|\hat{\Sigma}^O - \hat{\Sigma}\|_F^2. \end{aligned}$$

By adding the above two inequality, we obtain

$$\begin{aligned} &(\rho^- - \zeta^-) \|\hat{\Sigma} - \hat{\Sigma}^O\|_F^2 \\ &\leq \langle \nabla \tilde{f}(\hat{\Sigma}) - \nabla \tilde{f}(\hat{\Sigma}^O), \hat{\Sigma} - \hat{\Sigma}^O \rangle \\ &= \langle \nabla \tilde{f}(\hat{\Sigma}) + \lambda \hat{W}, \hat{\Sigma} - \hat{\Sigma}^O \rangle - \langle \nabla \tilde{f}(\hat{\Sigma}^O) + \lambda \hat{W}^O, \hat{\Sigma} - \hat{\Sigma}^O \rangle - \langle \lambda \hat{W} - \lambda \hat{W}^O, \hat{\Sigma} - \hat{\Sigma}^O \rangle \\ &= \underbrace{\langle \nabla \tilde{f}(\hat{\Sigma}) + \lambda \hat{W}, \hat{\Sigma} - \hat{\Sigma}^O \rangle}_I - \underbrace{\langle \nabla \tilde{f}(\hat{\Sigma}^O) + \lambda \hat{W}^O, \hat{\Sigma} - \hat{\Sigma}^O \rangle}_{III} \\ &\quad - \underbrace{\langle \nabla \tilde{f}(\hat{\Sigma}^O) + \lambda \hat{W}^O, \mathcal{P}^\perp(\hat{\Sigma}^O - \hat{\Sigma}) \rangle}_{III} - \underbrace{\langle \lambda \hat{W} - \lambda \hat{W}^O, \hat{\Sigma} - \hat{\Sigma}^O \rangle}_{IV}. \end{aligned}$$

Then we analysis the terms I, II, III and IV, respectively.

Since $\hat{\Sigma}$ is the optimal solution of (2), it satisfies the optimality condition. Therefore, we have

$$\langle \nabla \tilde{f}(\hat{\Sigma}) + \lambda \hat{W}, \hat{\Sigma} - \hat{\Sigma}^O \rangle \leq 0. \quad (31)$$

Based on the definition of the oracle estimator, the eigen-decompositions of $\hat{\Sigma}^O$ is $\hat{\Sigma}^O = \mathbf{U}^* \hat{\Gamma}^O \mathbf{U}^{*\top}$. Next, we decompose $\hat{\mathbf{W}}^O \in \partial \left\| \hat{\Sigma}^O \right\|_*$ as

$$\hat{\mathbf{W}}^O = \mathcal{P}(\hat{\mathbf{W}}^O) + \mathcal{P}^\perp(\hat{\mathbf{W}}^O) = \mathbf{U}^* \mathbf{U}^{*\top} + \hat{\mathbf{Z}}^O,$$

where $\hat{\mathbf{Z}}^O \in \mathcal{T}^\perp$ and $\left\| \hat{\mathbf{Z}}^O \right\|_2 \leq 1$. Since $\sum_{i=1}^d p_\lambda(\sigma_i(\Sigma)) = \lambda \|\Sigma\|_* + Q_\lambda(\Sigma)$, we have

$$\begin{aligned} \mathcal{P} \left(\nabla \left(\sum_{i=1}^d p_\lambda(\sigma_i(\hat{\Sigma}^O)) \right) \right) &= \mathcal{P} \left(\lambda \hat{\mathbf{W}}^O + \nabla Q_\lambda(\hat{\Sigma}^O) \right) \\ &= \mathcal{P} \left(\lambda (\mathbf{U}^* \mathbf{U}^{*\top} + \hat{\mathbf{Z}}^O) + \nabla Q_\lambda(\hat{\Sigma}^O) \right) \\ &= \mathcal{P} \left(\lambda \mathbf{U}^* \mathbf{U}^{*\top} + \nabla Q_\lambda(\hat{\Sigma}^O) \right) \\ &= \lambda \mathbf{U}^* \mathbf{U}^{*\top} + \mathbf{U}^* q'_\lambda \left((\hat{\Gamma}^O)_S \right) \mathbf{U}^{*\top} \\ &= \mathbf{U}^* \left(\lambda \mathbf{I} + q'_\lambda \left((\hat{\Gamma}^O)_S \right) \right) \mathbf{U}^{*\top}, \end{aligned} \quad (32)$$

and the diagonal matrix $\left(\lambda \mathbf{I} + q'_\lambda \left((\hat{\Gamma}^O)_S \right) \right)$ satisfies

$$\left[\lambda \mathbf{I} + q'_\lambda \left((\hat{\Gamma}^O)_S \right) \right]_{ii} = \lambda + q'_\lambda(\sigma_i(\hat{\Sigma}^O)) = p'_\lambda(\sigma_i(\hat{\Sigma}^O)), \quad (33)$$

where $i \in \mathcal{S}$. According to the perturbation bounds for eigenvalues, we have

$$\begin{aligned} \min_{i \in \mathcal{S}} |\sigma_i(\hat{\Sigma}^O)| &= \min_{i \in \mathcal{S}} |\sigma_i(\hat{\Sigma}^O) - \sigma_i^* + \sigma_i^*| \\ &\geq -\max_{i \in \mathcal{S}} |\sigma_i(\hat{\Sigma}^O) - \sigma_i^*| + \min_{i \in \mathcal{S}} |\sigma_i^*| \\ &\geq -\left\| \hat{\Sigma}^O - \Sigma^* \right\|_2 + \min_{i \in \mathcal{S}} |\sigma_i^*| \\ &\geq -\left\| \hat{\Sigma}^O - \Sigma^* \right\|_F + \min_{i \in \mathcal{S}} |\sigma_i^*| \\ &\geq -\frac{2\sqrt{r} \|\mathcal{P}(\nabla f(\Sigma^*))\|_2}{\rho^-} + \min_{i \in \mathcal{S}} |\sigma_i^*|. \end{aligned} \quad (34)$$

Since σ^* satisfies $\min_{i \in \mathcal{S}} |\sigma_i^*| \geq \alpha + \frac{2\sqrt{r} \|\mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma))\|_2}{m\rho^-} = \alpha + \frac{2\sqrt{r} \|\mathcal{P}(\nabla f(\Sigma^*))\|_2}{\rho^-}$, (34) can be updated to

$$\min_{i \in \mathcal{S}} |\sigma_i(\hat{\Sigma}^O)| \geq \alpha.$$

Given $\hat{\Sigma}^O \in \mathcal{T}$, we have $\text{rank}(\hat{\Sigma}^O) = r$, then $\sigma_i(\hat{\Sigma}^O)$ satisfies

$$\sigma_i(\hat{\Sigma}^O) \begin{cases} \geq \alpha & i \in \mathcal{S}, \\ = 0 & i \notin \mathcal{S}. \end{cases}$$

Under (d) in Assumption 1, for $i \in \mathcal{S}$, we have

$$p'_\lambda(\sigma_i(\hat{\Sigma}^O)) = 0. \quad (35)$$

Then combining (32), (33) and (35), we get

$$\mathcal{P} \left(\nabla \left(\sum_{i=1}^d p_\lambda(\sigma_i(\hat{\Sigma}^O)) \right) \right) = \mathbf{0}.$$

Since $\hat{\Sigma}^O$ is the oracle estimator, it satisfies the optimality condition. Therefore, we have the following inequality

$$\langle \nabla f(\hat{\Sigma}^O), \hat{\Sigma}^O - \hat{\Sigma} \rangle \leq 0.$$

Then we have that

$$\langle \nabla f(\hat{\Sigma}^O), \mathcal{P}(\hat{\Sigma}^O - \hat{\Sigma}) \rangle \leq \langle \nabla f(\hat{\Sigma}^O), \hat{\Sigma}^O - \hat{\Sigma} \rangle \leq 0.$$

Therefore, we have

$$\begin{aligned} & \langle \nabla \tilde{f}(\hat{\Sigma}^O) + \lambda \hat{W}^O, \mathcal{P}(\hat{\Sigma}^O - \hat{\Sigma}) \rangle \\ &= \langle \nabla f(\hat{\Sigma}^O), \mathcal{P}(\hat{\Sigma}^O - \hat{\Sigma}) \rangle + \left\langle \mathcal{P} \left(\nabla \left(\sum_{i=1}^d p_\lambda(\sigma_i(\hat{\Sigma}^O)) \right) \right), \mathcal{P}(\hat{\Sigma}^O - \hat{\Sigma}) \right\rangle \\ &\leq 0. \end{aligned} \tag{36}$$

By projecting $\nabla Q_\lambda(\hat{\Sigma}^O)$ into $\mathcal{T}^\perp$, we have

$$\begin{aligned} \mathcal{P}^\perp(\nabla Q_\lambda(\hat{\Sigma}^O)) &= (\mathbf{I} - \mathbf{U}^* \mathbf{U}^{*\top}) \mathbf{U}^* q'_\lambda(\hat{\mathbf{r}}^O) \mathbf{U}^{*\top} (\mathbf{I} - \mathbf{U}^* \mathbf{U}^{*\top}) \\ &= (\mathbf{U}^* - \mathbf{U}^*) q'_\lambda\left(\left(\hat{\mathbf{r}}^O\right)_{\bar{\mathcal{S}}}\right) (\mathbf{U}^* - \mathbf{U}^*) \\ &= \mathbf{0}. \end{aligned}$$

Under Assumption 4, it holds that

$$\begin{aligned} f(\Sigma^*) + \langle \nabla f(\Sigma^*), \hat{\Sigma}^O - \Sigma^* \rangle + \frac{\rho^+}{2} \|\hat{\Sigma}^O - \Sigma^*\|_F^2 &\geq f(\hat{\Sigma}^O). \\ f(\hat{\Sigma}^O) + \langle \nabla f(\hat{\Sigma}^O), \Sigma^* - \hat{\Sigma}^O \rangle + \frac{\rho^+}{2} \|\Sigma^* - \hat{\Sigma}^O\|_F^2 &\geq f(\Sigma^*). \end{aligned}$$

Adding the above two inequalities up, we get

$$\begin{aligned} \rho^+ \|\hat{\Sigma}^O - \Sigma^*\|_F^2 &\geq \langle \nabla f(\hat{\Sigma}^O) - \nabla f(\Sigma^*), \hat{\Sigma}^O - \Sigma^* \rangle \\ &\geq \|\nabla f(\hat{\Sigma}^O) - \nabla f(\Sigma^*)\|_F \cdot \|\hat{\Sigma}^O - \Sigma^*\|_F, \end{aligned}$$

which can be simplified as

$$\|\nabla f(\hat{\Sigma}^O) - \nabla f(\Sigma^*)\|_F \leq \rho^+ \|\hat{\Sigma}^O - \Sigma^*\|_F.$$

Thus, we have

$$\begin{aligned} \|\mathcal{P}^\perp(\nabla f(\hat{\Sigma}^O))\|_2 &\leq \|\nabla f(\hat{\Sigma}^O)\|_2 \\ &\leq \|\nabla f(\hat{\Sigma}^O) - \nabla f(\Sigma^*)\|_2 + \|\nabla f(\Sigma^*)\|_2 \\ &\leq \|\nabla f(\hat{\Sigma}^O) - \nabla f(\Sigma^*)\|_F + \|\nabla f(\Sigma^*)\|_2 \\ &\leq \rho^+ \|\hat{\Sigma}^O - \Sigma^*\|_F + \|\nabla f(\Sigma^*)\|_2 \\ &\leq \frac{2\rho^+ \sqrt{r} \|\mathcal{P}(\nabla f(\Sigma^*))\|_2}{\rho^-} + \|\nabla f(\Sigma^*)\|_2 \\ &\leq \frac{2\rho^+ \sqrt{r} \|\nabla f(\Sigma^*)\|_2}{\rho^-} + \|\nabla f(\Sigma^*)\|_2 \\ &= \frac{2\rho^+ \sqrt{r} \|\mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma))\|_2}{m\rho^-} + \frac{\|\mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma))\|_2}{m}. \end{aligned}$$

Under the condition that $\lambda \geq 2m^{-1} \|\mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}))\|_2 + 2m^{-1} \sqrt{r} \rho^+ \|\mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}))\|_2 / \rho^-$, the upper bound of $\|\mathcal{P}^\perp(\nabla f(\hat{\boldsymbol{\Sigma}}^O))\|_2$ satisfies

$$\|\mathcal{P}^\perp(\nabla f(\hat{\boldsymbol{\Sigma}}^O))\|_2 \leq \lambda.$$

So we can decompose $\hat{\mathbf{W}}^O \in \partial \|\hat{\boldsymbol{\Sigma}}^O\|_*$ as

$$\hat{\mathbf{W}}^O = \mathcal{P}(\hat{\mathbf{W}}^O) + \mathcal{P}^\perp(\hat{\mathbf{W}}^O) = \mathbf{U}^* \mathbf{U}^{*\top} - \lambda^{-1} \mathcal{P}^\perp(\nabla f(\hat{\boldsymbol{\Sigma}}^O)).$$

Then we get

$$\begin{aligned} & \langle \nabla \tilde{f}(\hat{\boldsymbol{\Sigma}}^O) + \lambda \hat{\mathbf{W}}^O, \mathcal{P}^\perp(\hat{\boldsymbol{\Sigma}}^O - \hat{\boldsymbol{\Sigma}}) \rangle \\ &= \langle \mathcal{P}^\perp(\nabla f(\hat{\boldsymbol{\Sigma}}^O) + \lambda \hat{\mathbf{W}}^O), \mathcal{P}^\perp(\hat{\boldsymbol{\Sigma}}^O - \hat{\boldsymbol{\Sigma}}) \rangle + \langle \mathcal{P}^\perp(\nabla Q_\lambda(\hat{\boldsymbol{\Sigma}}^O)), \mathcal{P}^\perp(\hat{\boldsymbol{\Sigma}}^O - \hat{\boldsymbol{\Sigma}}) \rangle \\ &= \langle \mathbf{0}, \mathcal{P}^\perp(\hat{\boldsymbol{\Sigma}}^O - \hat{\boldsymbol{\Sigma}}) \rangle + \langle \mathbf{0}, \mathcal{P}^\perp(\hat{\boldsymbol{\Sigma}}^O - \hat{\boldsymbol{\Sigma}}) \rangle \\ &= 0. \end{aligned} \tag{37}$$

Additionally, since $\|\cdot\|_*$ is convex, $\hat{\mathbf{W}} \in \partial \|\hat{\boldsymbol{\Sigma}}\|_*$ and $\hat{\mathbf{W}}^O \in \partial \|\boldsymbol{\Sigma}^*\|_*$, we have

$$\langle \lambda \hat{\mathbf{W}} - \lambda \hat{\mathbf{W}}^O, \hat{\boldsymbol{\Sigma}} - \hat{\boldsymbol{\Sigma}}^O \rangle \geq 0. \tag{38}$$

Therefore, by adding up (31), (36), (37) and (38), we obtain

$$(\rho^- - \zeta^-) \|\hat{\boldsymbol{\Sigma}} - \hat{\boldsymbol{\Sigma}}^O\|_F^2 \leq 0.$$

Since $\rho^- > \zeta^-$, it follows that $\hat{\boldsymbol{\Sigma}}^O = \hat{\boldsymbol{\Sigma}}$. Thus, we have

$$\|\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^*\|_F = \|\hat{\boldsymbol{\Sigma}}^O - \boldsymbol{\Sigma}^*\|_F \leq \frac{2\sqrt{r} \|\mathcal{P}(\nabla f(\boldsymbol{\Sigma}^*))\|_2}{\rho^-}$$

□

A.4 PROOF OF COROLLARY 9

First, we give three essential Lemmas.

Lemma 12 (Proposition 1 in Negahban and Wainwright [2011]). *Consider the sampling operator of $\boldsymbol{\Theta}$ -ensemble, it holds with probability at least $1 - 2 \exp(-m/32)$ that*

$$\frac{\|\mathcal{A}(\boldsymbol{\Delta})\|_2}{\sqrt{m}} \geq \frac{1}{4} \left\| \sqrt{\boldsymbol{\Theta}} \text{vec}(\boldsymbol{\Delta}) \right\|_2 - 24\pi(\boldsymbol{\Theta}) \sqrt{d/m} \|\boldsymbol{\Delta}\|_*.$$

Lemma 13 (Lemma 6 in Negahban and Wainwright [2011]). *There're universal constants C_0, C_1 and C_2 such that*

$$\mathbb{P} \left[\frac{\|\mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}))\|_2}{m} \geq 2C_0 \delta \varpi \sqrt{d/m} \right] \leq C_1 \exp(-2dC_2).$$

Lemma 14 (Lemma D.1 in Sun et al. [2018]). *Let $\mathbf{x}$ be a sub-Gaussian random vector with zero mean and covariance $\boldsymbol{\Sigma}^*$ and $\{\mathbf{x}_i\}_{i=1}^n$ be a collection of i.i.d. samples from $\mathbf{x}$. There exists some constants c_1, c_2 , and z_0 such that for all z with $0 < z < z_0$, the sample covariance matrix $\mathbf{S}$ satisfies the following tail bound*

$$\mathbb{P}(|\Sigma_{ij}^* - S_{ij}| > z) \leq c_1 \exp(-c_2 n z^2).$$

Then, we give the proof of Corollary 9.

Proof. By projecting $\|\hat{\Delta}\|_*$ into $\mathcal{T}$ and $\mathcal{T}^\perp$, we have

$$\|\hat{\Delta}\|_* = \|\mathcal{P}(\hat{\Delta})\|_* + \|\mathcal{P}^\perp(\hat{\Delta})\|_* \leq 6\|\mathcal{P}(\hat{\Delta})\|_* = 6\sqrt{r}\|\mathcal{P}(\hat{\Delta})\|_F \leq 6\sqrt{r}\|\hat{\Delta}\|_F.$$

Then according to Lemma 12, we have

$$\begin{aligned} \frac{\|\mathcal{A}(\hat{\Delta})\|_2}{\sqrt{m}} &\geq \frac{1}{4}\|\sqrt{\Theta}\text{vec}(\hat{\Delta})\|_2 - 24\pi(\Theta)\sqrt{d/m}\|\hat{\Delta}\|_* \\ &\geq \frac{\sqrt{\gamma_{\min}(\Theta)}}{4}\|\hat{\Delta}\|_F - 144\sqrt{r}\pi(\Theta)\sqrt{d/m}\|\hat{\Delta}\|_F. \end{aligned}$$

Then for $m \geq \frac{C_1 r \pi^2(\Theta) d^2}{\gamma_{\min}(\Theta)}$ where C_1 is sufficiently large such that

$$144\sqrt{r}\pi(\Theta)\sqrt{d/m} \leq \frac{\sqrt{\gamma_{\min}(\Theta)}}{8},$$

we have

$$\frac{\|\mathcal{A}(\hat{\Delta})\|_2}{\sqrt{m}} \geq \frac{\sqrt{\gamma_{\min}(\Theta)}}{8}\|\hat{\Delta}\|_F,$$

which implies that

$$\frac{\|\mathcal{A}(\hat{\Delta})\|_2^2}{m} \geq \frac{\gamma_{\min}(\Theta)}{64}\|\hat{\Delta}\|_F^2,$$

for $\rho^- = C_2\gamma_{\min}(\Theta)$, the above inequality can be updated to

$$\frac{\|\mathcal{A}(\hat{\Delta})\|_2^2}{m} \geq \frac{\rho^-}{64C_2}\|\hat{\Delta}\|_F^2,$$

when we set $C_2 = \frac{1}{32}$, the Restricted Strong Convexity of $\mathcal{A}(\cdot)$ is established. Additionally, we have

$$\|\mathcal{P}_{\mathcal{S}_1}(\nabla f(\Sigma))\|_2 = \left\|U_{\mathcal{S}_1}^{\star\top} \nabla f(\Sigma) U_{\mathcal{S}_1}^*\right\|_2.$$

According to Lemma 13, since $U_{\mathcal{S}_1}^{\star\top} \nabla f(\Sigma) U_{\mathcal{S}_1}^* \in \mathbb{R}^{d \times d}$, combining with Lemma 14, we have

$$\begin{aligned} \left\|U^{\star\top} \nabla f(\Sigma) U^*\right\|_2 &\leq 2C_0\delta\varpi\sqrt{d/(mn)}, \\ \left\|U_{\mathcal{S}_1}^{\star\top} \nabla f(\Sigma) U_{\mathcal{S}_1}^*\right\|_2 &\leq 2C_0\delta\varpi\sqrt{|\mathcal{S}_1|/(mn)}, \end{aligned}$$

which holds with probability at least $1 - C_1 \exp(-2|\mathcal{S}_1|C_2)$. Based on Theorem 6, since $\vartheta = \left\|\mathcal{P}_{\mathcal{S}_1}(\nabla \tilde{f}(\Sigma^*))\right\|_2$, $\lambda = \|\nabla f(\Sigma^*)\|_2$ and $\rho^- = C_2\gamma_{\min}(\Theta)$

$$\begin{aligned} \left\|\hat{\Sigma} - \Sigma^*\right\|_F &\leq \frac{2\vartheta\sqrt{|\mathcal{S}_1|}}{\rho^-} + \frac{6\lambda\sqrt{|\mathcal{S}_2|}}{\rho^-} \\ &\leq \frac{1}{C_2\gamma_{\min}(\Theta)} \left(4C_0\delta\varpi\sqrt{|\mathcal{S}_1|/(mn)} \cdot \sqrt{|\mathcal{S}_1|} + 12C_0\delta\varpi\sqrt{d/(mn)} \cdot \sqrt{|\mathcal{S}_2|}\right) \\ &\leq \frac{\delta\varpi}{\gamma_{\min}(\Theta)\sqrt{mn}} \left(4C_0|\mathcal{S}_1|/C_2 + 12C_0\sqrt{d}/C_2\right), \end{aligned}$$

which holds with probability at least $1 - C_1 \exp(-2|\mathcal{S}_1|C_2)$. Therefore, by choosing appropriate value of C_0, C_1, C_2, C_5 and C_6 , we get

$$\left\|\hat{\Sigma} - \Sigma^*\right\|_F \leq \frac{\delta\varpi}{\sqrt{mn}\gamma_{\min}(\Theta)} \left[C_1|\mathcal{S}_1| + C_2\sqrt{|\mathcal{S}_2|d}\right],$$

which holds with probability at least $1 - C_5 \exp(-2dC_6)$. Since $\rho^- = C_1 \gamma_{\min}(\boldsymbol{\Theta})$ and Lemma 13, according to Corollary 8, we have

$$\begin{aligned} \min_{i \in \mathcal{S}} |\sigma_i^*| &\geq \alpha + \frac{2\sqrt{r} \|\sum_{i=1}^m \eta_i \mathbf{A}_i\|_2}{m\rho^-} \\ &\geq \alpha + \frac{C_2 \delta \varpi \sqrt{rd/m}}{\gamma_{\min}(\boldsymbol{\Theta})} \\ &\geq \alpha + \frac{C_2 m^{-1} \|\sum_{i=1}^m \eta_i \mathbf{A}_i\|_2 \sqrt{r}}{2\rho^-}, \end{aligned}$$

which holds with probability at least $1 - C_1 \exp(-2dC_2)$, where $\mathbf{A}_i := \mathbf{a}_i \mathbf{a}_i^\top$. With suitable value of C_2 , the above inequality is equal to

$$\min_{i \in \mathcal{S}} |\sigma_i^*| \geq \alpha + \frac{2\sqrt{r} \|\sum_{i=1}^m \eta_i \mathbf{A}_i\|_2}{m\rho^-},$$

which holds with probability at least $1 - C_1 \exp(-2dC_2)$ and is the requirement of $\boldsymbol{\sigma}^*$ in Corollary 8. According to the definition of $f(\cdot)$, we have

$$\mathbf{H}_m = \frac{\sum_{i=1}^m \text{vec}(\mathbf{A}_i) \text{vec}(\mathbf{A}_i)^\top}{m},$$

where $\mathbf{H}_m$ is the Hessian matrix of $f(\cdot)$. Then we have $\mathbb{E}(\mathbf{H}_m) = \boldsymbol{\Theta}$.

By concentration, when m is sufficiently large, with high probability, $\gamma_{\max}(\mathbf{H}_m) \leq 2\gamma_{\max}(\boldsymbol{\Theta})$. Based on 4, we could obtain that $\rho^+ = \gamma_{\max}(\mathbf{H}_m)$. Then we have that when m is sufficiently large, with high probability, $\rho^+ \leq 2\gamma_{\max}(\boldsymbol{\Theta})$. Additionally, given $\rho^- = C_1 \gamma_{\min}(\boldsymbol{\Theta})$ and Lemma 13, we have

$$\begin{aligned} \lambda &\geq C_3 \delta \varpi \sqrt{d/m} \left(1 + \frac{\sqrt{r} \gamma_{\max}(\boldsymbol{\Theta})}{\gamma_{\min}(\boldsymbol{\Theta})} \right) \\ &\geq C_3 \delta \varpi \sqrt{d/m} \left(1 + \frac{\sqrt{r} \rho^+ C_1}{2\rho^-} \right) \\ &\geq \frac{C_3}{2C_1} m^{-1} \left\| \sum_{i=1}^m \eta_i \mathbf{A}_i \right\|_2 \left(1 + \frac{\sqrt{r} \rho^+ C_1}{2\rho^-} \right), \end{aligned}$$

which holds with probability at least $1 - C_1 \exp(-2dC_2)$. With suitable value of C_1 and C_3 , the above inequality is equal to

$$\lambda \geq 2m^{-1} \left\| \sum_{i=1}^m \eta_i \mathbf{A}_i \right\|_2 + 2m^{-1} \sqrt{r} \rho^+ \left\| \sum_{i=1}^m \eta_i \mathbf{A}_i \right\|_2 / \rho^-,$$

which holds with probability at least $1 - C_1 \exp(-2dC_2)$ and is the requirement of λ in Corollary 8. Since the requirements of $\boldsymbol{\sigma}^*$ and λ are reached with probability at least $1 - C_1 \exp(-2dC_2)$, according to Corollary 8, we have that $\hat{\boldsymbol{\Sigma}}$ is the optimal solution of the estimator, $\text{rank}(\hat{\boldsymbol{\Sigma}}) = \text{rank}(\boldsymbol{\Sigma}^*) = r$, with probability at least $1 - C_4 \exp(-C_5 d)$ by setting the proper values of C_4 and C_5 .

In addition, we have

$$\begin{aligned} \left\| \hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}^* \right\|_{\text{F}} &\leq \frac{2\sqrt{r} \left\| \mathcal{P}_{S_1}(\nabla \tilde{f}(\boldsymbol{\Sigma}^*)) \right\|_2}{\rho^-} \\ &\leq \frac{2\sqrt{r} \left\| \mathcal{P}_{S_1}(\nabla \tilde{f}(\boldsymbol{\Sigma}^*)) \right\|_2}{C_1 \gamma_{\min}(\boldsymbol{\Theta})} \\ &\leq \frac{4\sqrt{r} C_0 \delta \varpi \sqrt{r/(mn)}}{C_1 \gamma_{\min}(\boldsymbol{\Theta})} \\ &\leq \frac{4C_0 r \delta \varpi}{C_1 \sqrt{mn} \gamma_{\min}(\boldsymbol{\Theta})} \\ &\leq \frac{C_6 r \delta \varpi}{\sqrt{mn} \gamma_{\min}(\boldsymbol{\Theta})}, \end{aligned}$$

which holds with probability at least $1 - C_4 \exp(-C_5 d)$ with suitable value of C_1, C_2, C_4, C_5 and C_6 .

□